

MFE 230GA · FINAL PROJECT · FALL 2026

# Follow the Workers

Do labor-market flows predict which U.S. industries outperform?

---

## TEAM

|          |         |
|----------|---------|
| Piero    | PELOSI  |
| Alex     | ROESLER |
| Romain   | ALMEIDA |
| Elouan   | BAHRI   |
| Al Yazid | BENSAID |

## COURSE

MFE 230GA – Equity Markets

## INSTRUCTORS

Raffaele Savi and Gerald Garvey  
BlackRock

## GRADUATE STUDENT INSTRUCTOR

Vinicio DeSola

## 1 Executive summary

---

We propose a market-neutral industry-rotation strategy: rank 13 U.S. industry groups each month on the year-over-year change in openings, hires, layoffs and hours, as known at the time, and hold the top three against the bottom three. It was preregistered (Study 1) and tested once on 142 months, November 2014 to August 2026. It lost 4.6% a year after costs (Sharpe  $-0.58$ ); its forecasts were uncorrelated with returns and it failed four of five preset criteria. We conclude: *Do not implement*. Afterwards we found two of its four signs opposite to the data in both periods, its one significant signal, wage growth, dropped, and its AI-proposed features switched off in almost every test month. In a post-hoc iteration (Study 2) the data are consistent with a slower, opposite effect that investment-based asset pricing predicts: industries that hire and lose workers fastest earn lower returns over the next 9 to 24 months. A 12-month book earns a test Sharpe of  $+0.29$  ( $+0.20$  on data as known at the time), but it was designed after the test was opened, lost money in the first test half and does not survive a correction for the looks taken. It runs as a paper forward test frozen 3 October 2026. Claude was a useful auditor and idea generator when a human checked every claim and it could not change the rules after seeing results.

## 2 What did we try? Labor flows as industry news

---

### 2.1 Ideas: demand news, cost news and tightness

*Origin.* The idea is a macro driver turned into a cross-sectional signal. The labor market reports an industry's prospects before its earnings do: the Bureau of Labor Statistics (BLS) publishes job openings, hires, quits and layoffs by industry every month in the Job Openings and Labor Turnover Survey (JOLTS), and hours and earnings in the Current Employment Statistics (CES). We turn each series into an industry score and judge it by its information coefficient (IC), the rank correlation between the score and the next relative return. The benchmark it has to beat is industry momentum, the signal behind most published industry rotation (Moskowitz and Grinblatt, 1999; Zarattini and Antonacci, 2024). We state three hypotheses (Figure 1).

- *H1, demand channel* (preregistered and tested). Rising openings, hires and hours signal growing demand for an industry's output before it reaches reported sales, so the industry outperforms over the next one to three months (Hong et al., 2007).
- *H2, cost and investment channel* (tested post hoc). Firms that hire heavily convert growth options into assets in place and pay adjustment costs, so their expected returns fall over the following 6 to 24 months (Belo et al., 2014); wage pressure squeezes margins, and labor-market frictions make labor-intensive firms riskier (Kuehn et al., 2017; Donangelo et al., 2019).
- *H3, tightness moderator*. When vacancies outnumber the unemployed, hiring is expensive and H2 should dominate H1. We measure tightness by  $T_t = \log(\text{job openings}_t / \text{unemployed}_t)$ ;  $T > 0$  means more vacancies than unemployed ( $V/U > 1$ ).

*Two studies.* Study 1 traded H1: a monthly book long the industries with rising openings, hires and hours and falling layoffs, preregistered and tested once. It is the only clean out-of-sample test in this paper. Study 2 starts from its failure. The diagnosis (Section 3) showed that the one-month signs were wrong or absent and that hiring and quits predict returns with the opposite sign at long horizons, which is H2. Study 2 asks whether a slow cost-channel book works, with four variants written down on 3 October 2026 (Table 3). The test period had been opened by then, so every Study 2 number carries the post-hoc label, and the only clean evidence H2 can still earn is the forward test.

*Why Study 1 carries the paper.* (a) It is the only clean out-of-sample test we have; it was run once and we report it as it came out. (b) Its failure produced the candidate: the diagnosis in Section 3 pointed to wage growth and to the 12-month hiring reversal, and Study 2 builds on it. (c) It is the reason Study 2 must be labelled post hoc and the reason the forward test exists. (d) It contains the agent experiment, which is the original part of the design and the basis of our evaluation of Claude (Section 3.4).

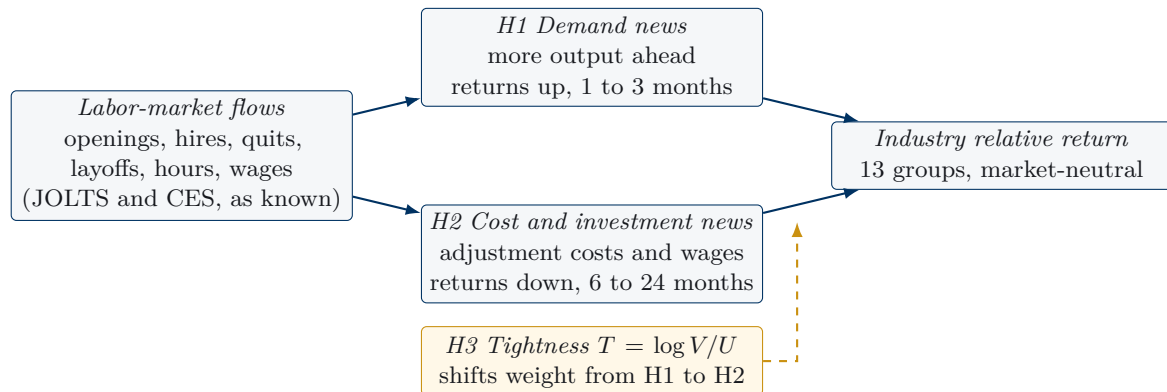


FIGURE 1. The three hypotheses. Labor flows carry two kinds of industry news with opposite sign and different horizons (H1, H2); tightness should change which one dominates (H3).

TABLE 1. Signals, preregistered signs and what the data say. **CONFIRMATORY** (design) / **POST-HOC** (last column)

| Signal      | Definition (group $g$ , month $t$ )              | Prior   | Mechanism                                       | IC, $h=1$ |
|-------------|--------------------------------------------------|---------|-------------------------------------------------|-----------|
| F1 openings | 12-month change in 3-month average openings rate | +       | H1 demand news                                  | −0.005    |
| F2 hires    | same, hires rate                                 | +       | H1 demand news; H2 at long horizons             | +0.013    |
| F3 quits    | same, quits rate                                 | none    | H2, worker bargaining power                     | +0.009    |
| F4 layoffs  | same, layoffs rate                               | −       | H1, negative demand shock                       | +0.028    |
| F5 hours    | 12-month log change, average weekly hours        | +       | H1, intensive margin                            | −0.010    |
| W wages     | 12-month log change, average hourly earnings     | dropped | H2 cost pressure (−) vs. labor risk premium (+) | +0.039    |

IC is the average monthly Spearman correlation between the signal and the next month's industry relative return, May 2004 to August 2026, revised data with publication lags. Throughout,  $t$ -statistics use Newey–West standard errors, which allow for autocorrelation (HAC); only W has  $|t| > 2$  ( $t = 2.50$ ).

## 2.2 Data: point-in-time labor flows and industry returns

All sources are free and on the brief's suggested list: the Ken French Data Library, FRED and BLS employment data (Table 2; links and access dates in Appendix G).

*Four data versions.* Labor data are revised after first release, so today's values leak information. Study 1 rebuilt every series as it stood on each decision date from ALFRED, the St. Louis Fed archive of data vintages (78 series), and kept three alternatives: first release, fully revised, and revised with fixed publication lags. Its result barely depends on the version (Table 8). Study 2 first used revised data with fixed lags, under which the fixed rule A0 picks the same top-three groups as on as-known data in only 42% of months; Study 2.3 therefore reran every post-hoc variant on as-known data from Study 1's saved panel, after checking that our rebuild reproduces its features exactly (Table 9). The rerun also removes a flaw of the fixed-lag rule, which assumes the normal release calendar: the federal shutdown of October to November 2025 delayed releases, so two of the last 18 months of the fixed-lag Study 2 numbers use data published after the decision.

TABLE 2. Data sources.

CONFIRMATORY

| Source     | Series                                                                                   | Freq. | Known at $t$ | Vintage handling in Study 1                            |
|------------|------------------------------------------------------------------------------------------|-------|--------------|--------------------------------------------------------|
| BLS JOLTS  | openings, hires, quits, layoffs rates; 13 industries; not seasonally adjusted            | M     | $t - 2$      | ALFRED vintages: the value known on each decision date |
| BLS CES    | average weekly hours and hourly earnings, production and nonsupervisory workers          | M     | $t - 1$      | as above                                               |
| FRED       | total openings (JTSJOL) and unemployed (UNEMPLOY) for $T_t$                              | M     | $t - 2$      | as above                                               |
| Ken French | 49 value-weighted industry portfolios $\rightarrow$ 13 groups; five factors and momentum | M     | $t$          | CRSP build 202608                                      |

*Universe and effective breadth.* The 49 French industries are mapped to the 13 JOLTS/CES industry groups and value-weighted within each group (Appendix H). The groups are not 13 independent bets: the participation ratio of the correlation matrix of their relative returns,  $(\sum \lambda)^2 / \sum \lambda^2$  over its eigenvalues, is 9.1. By the fundamental law of active management the information ratio is about  $IC \times \sqrt{BR} \times TC$ , where BR is breadth (independent bets a year) and TC the transfer coefficient (the correlation between the signal and the weights held). With 9 effective groups a monthly IC of 0.05 caps the ratio near 0.5 and a 12-month IC of 0.10 near 0.3. Even with all 13 groups counted as independent, the law implies only +0.17 for the 12-month book of Study 2 (TC 0.48), against a realised +0.29. The mapping is also coarse: labor data describe every establishment in an industry, returns describe listed firms, and durable manufacturing is 60% semiconductors by market capitalisation.

*Sample.* Research (signal design and model selection): May 2004 to September 2014, 125 months. Test, opened once: November 2014 to August 2026, 142 months. Figure 2 shows why H3 could not be learned:  $V/U$  never exceeded 1 in the research period and exceeded it in 75 test months.

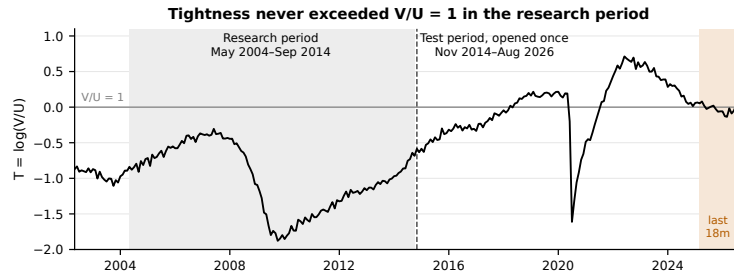


FIGURE 2. Tightness never exceeded  $V/U = 1$  in the research period, so a moderator learned there (H3) is extrapolated throughout the test.

POST-HOC

### 2.3 Implementation: signals, models, risk, costs and AI prompts

*Signals.* For a labor series  $x_{g,t}$  in group  $g$ , let  $\bar{x}_{g,t} = \frac{1}{3}(x_{g,t} + x_{g,t-1} + x_{g,t-2})$ . With publication lag  $\ell$  (2 months for JOLTS, 1 for CES),

$$F_{g,t} = \bar{x}_{g,t-\ell} - \bar{x}_{g,t-\ell-12} \quad (\text{log differences for hours and earnings}). \quad (1)$$

Each signal is divided by its own expanding-window standard deviation (at least 24 months), z-scored across the 13 groups and winsorised at  $\pm 3$ , giving  $z_{g,t}^{(k)}$ . Year-over-year differences remove seasonality.

*Models.* Table 3 lists every strategy. The learned ones in Study 1 are ridge regressions of next-month relative return on the signals,

$$\hat{\beta}_t = \arg \min_{\beta} \sum_{s < t} \sum_g (r_{g,s+1}^{\text{rel}} - z_{g,s}^{\top} \beta)^2 + \alpha \|\beta\|^2, \quad (2)$$

re-estimated each month on an expanding window, with  $\alpha$  chosen once by cross-validation with a gap between folds, inside the research period; it chose 1000, the strongest shrinkage on the grid, for every model.

TABLE 3. Strategies and their names.

CONFIRMATORY (Study 1) / POST-HOC (Study 2)

|                                                                                        |                               |                                                                                             |
|----------------------------------------------------------------------------------------|-------------------------------|---------------------------------------------------------------------------------------------|
| <i>Study 1, preregistered (Alex): H1 books, one-month horizon, three-month holding</i> |                               |                                                                                             |
| A0                                                                                     | fixed rule                    | $S_{g,t} = \frac{1}{4}(z^{(1)} + z^{(2)} - z^{(4)} + z^{(5)})$ , no estimation              |
| A1                                                                                     | ridge                         | ridge on F1–F5                                                                              |
| A1-T                                                                                   | ridge $\times$ tightness      | A1 plus each signal interacted with $T$ (H3)                                                |
| A3                                                                                     | agent features, independent   | A1 plus features proposed by independent LLM agents; the primary, chosen on research data   |
| A4                                                                                     | agent features, communicating | A1 plus features proposed by agents that exchange reports                                   |
| <i>Study 2, post-hoc iteration (specs hashed before running)</i>                       |                               |                                                                                             |
| W                                                                                      | wage growth                   | cross-sectional rank of $z(W)$ ; the declared headline                                      |
| SC                                                                                     | signed composite              | $\frac{1}{6} \sum_k \text{sign}_k z^{(k)}$ over F1–F5 and W, signs learned on research data |
| W+MOM                                                                                  | wage growth + momentum        | average rank of $z(W)$ and 12-month industry momentum                                       |
| HQ12                                                                                   | hires + quits reversal        | $-\frac{1}{2}(z(F2) + z(F3))$ , held 12 months; the H2 book                                 |

*How Study 2 was run.* Each stage was written to a specification file, hashed (a fingerprint that changes if the file does) and committed before it ran, then run once; a run log stores the hash of every output, and the notebook stops if a later run differs (Table 27, appendix). Study 2.1 fixed the four variants and a reading rule, Study 2.2 a battery of seven pre-declared conditions for HQ12, and Study 2.3 the as-known rerun, the cap and rank-weight variants and the tests of the agent experiment. Every new backtest counts as one more look in the deflated Sharpe ratio.<sup>1</sup> All of them use Study 1’s own backtest code, which our notebook reproduces to  $10^{-16}$  before anything else runs.

*Risk modeling.* Each month the score  $s_{m,g}$  known at the end of month  $m - 1$  becomes a dollar-neutral tranche, long the three highest-ranked groups and short the three lowest; the book averages the last three tranches (twelve for HQ12). The risk model is a Ledoit–Wolf shrinkage covariance of the 13 groups and the market over 60 months (Ledoit and Wolf, 2004). The book is hedged to zero market beta with 60-month betas, scaled to a 10% ex-ante volatility and capped at a gross exposure of 2 (the equations are in Appendix B). The cap, not the volatility target, was the binding constraint: thirteen correlated, hedged groups carry so little risk per unit of gross that reaching 10% would need a median gross exposure of 2.6 for A3 and 2.4 for HQ12. Table 4 shows that lifting the cap restores the volatility and leaves the Sharpe ratio where it was, so the cap explains the volatility shortfall and not the losses.

*Turnover and costs.* Every book pays 10 bp per unit of industry turnover and 2 bp on the hedge. Table 5 gives annual turnover, the cost drag and the break-even cost. A3 turns its book over eleven times a year and loses money even with free trading; HQ12 trades 4.4 times a year and breaks even at 68 bp. Holding all 13 groups in proportion to their rank instead of the top and bottom three changes little: A3 moves from  $-0.58$  to  $-0.49$  and HQ12 from  $+0.29$  to  $+0.31$ . Capacity is small with the liquid proxies we would trade (one ETF per group): the least liquid one trades \$3.2 million a day, so monthly trading reaches 1% of a day’s volume at about \$1.2 million of assets for HQ12 and \$0.5 million for A3. The implementability criterion (G5 below) stays failed.

<sup>1</sup>Each table’s deflated Sharpe ratio uses the looks taken when that table was run: 112 for Study 2.1 (the four variants), 132 for Study 2.2 (the HQ12 battery) and 162 for Study 2.3 (the as-known, cap and rank-weighted runs). Later counts are larger by design.

TABLE 4. Risk overlay in the test window: A3 and HQ12 with the gross cap at 2 (as run), 3 and none. **POST-HOC** (diagnostic)

| Book | Cap  | Sharpe | Realised vol | Mean gross | Cap binds | Max drawdown |
|------|------|--------|--------------|------------|-----------|--------------|
| A3   | 2    | −0.58  | 7.8%         | 1.98       | 87%       | −41.1%       |
| A3   | 3    | −0.61  | 9.7%         | 2.57       | 35%       | −50.7%       |
| A3   | none | −0.62  | 10.3%        | 2.79       | 0%        | −54.9%       |
| HQ12 | 2    | +0.29  | 8.7%         | 1.97       | 83%       | −14.4%       |
| HQ12 | 3    | +0.24  | 10.5%        | 2.51       | 39%       | −20.3%       |
| HQ12 | none | +0.24  | 11.1%        | 2.78       | 0%        | −22.6%       |

TABLE 5. Turnover and costs in the test window. Break-even is the industry cost at which net return is zero. **CONFIRMATORY** (A3) / **POST-HOC**

| Book (test window)                         | Turnover p.a. | Cost drag p.a. | Gross p.a. | Net p.a. | Break-even (bp) |
|--------------------------------------------|---------------|----------------|------------|----------|-----------------|
| A3 agent features, independent (primary)   | 11.0x         | 1.11%          | −3.5%      | −4.6%    | −32             |
| W wage growth (declared headline)          | 4.7x          | 0.47%          | +2.1%      | +1.7%    | +46             |
| SC signed composite                        | 8.7x          | 0.88%          | −0.5%      | −1.3%    | −5              |
| W+MOM wage growth + momentum               | 5.8x          | 0.59%          | +3.6%      | +3.1%    | +62             |
| HQ12 hires + quits reversal, 12-month hold | 4.4x          | 0.44%          | +3.0%      | +2.6%    | +68             |
| A3 rank-weighted                           | 10.3x         | 1.05%          | −2.5%      | −3.6%    | −24             |
| HQ12 rank-weighted                         | 4.4x          | 0.45%          | +3.0%      | +2.6%    | +69             |

#### HOW WE KEPT THE TEST HONEST

- Every input as it was known on the decision date; the primary result is unchanged under four data versions.
- Signals, models, the primary strategy and five pass/fail gates (G1 timing, G2 prediction, G3 incremental alpha, G4 robustness, G5 implementability) were fixed and hashed before any test data were touched, and the test was opened once. We verified the record: the hash of the freeze file equals the one stored when the test was opened, 0.04 seconds after the freeze, and 337 of the 338 frozen files in the repository still match their hashes (the exception is the human review written afterwards).
- Every result is reported whatever its sign, with a placebo, drop-one-group checks, comparisons corrected for multiple testing (Holm) and a deflated Sharpe ratio (DSR), which discounts a Sharpe ratio for the number of looks taken (Bailey and López de Prado, 2014).
- Anything decided after seeing test results is labelled **POST-HOC** and was itself written to a file and hashed before it ran; the paper book is labelled **FORWARD TEST**.

*AI prompts.* We used Claude (Anthropic) for the three uses the brief names. For *idea generation*, blinded agents proposed new labor features from six anonymised variables in a restricted formula grammar: two arms of three agents, independent and communicating, three seeds, 108 proposals per study, with mechanical selection and a human-audited judge (design diagram and results in Appendix B). For *coding support*, Claude audited the data mapping and the as-of code before any result existed. For *robustness design*, it red-teamed the research results. The three prompts are below (P2 is the opening of its system prompt); outputs and our evaluation are in Section 3.4, transcripts in Appendix E.1.

#### IDEA GENERATION. PROMPT P2: BLINDED SIGNAL AGENT (SYSTEM PROMPT, OPENING)

You are a quantitative researcher. You will propose one feature per turn to predict which of 13 anonymous groups (G01 to G13) will have higher returns next month relative to the others. You see only the variable dictionary below, the allowed operations, and summary statistics of your previous submissions. You do not know what the groups, variables or time periods are. Do not guess them and do not use knowledge about any real industry, company or date. [variable dictionary, grammar and JSON format follow]

## CODING SUPPORT. PROMPT P1: DATA AND CODE AUDIT

Find mapping errors, timestamp or look-ahead errors, and code bugs in the material below. For each problem, give the exact line or row, why it is wrong, and a test that would confirm it. Do not report style issues. [followed by the series manifest, the industry crosswalk and the as-of code]

## ROBUSTNESS DESIGN. PROMPT P4: RED TEAM

Give 10 distinct reasons these results could be spurious, ranked by likelihood. For each, name a test that can be run with this data and the result that would make us say "do not implement". [followed by the research-period result tables, without dates or names]

### 3 What did we learn? Slow cost news, not fast demand news

## ANSWER TO THE THESIS

- *H1, demand channel: rejected* **CONFIRMATORY**. The primary H1 book (A3) lost 4.6% a year (Sharpe  $-0.58$ ), its IC was  $-0.021$  ( $t = -1.08$ ), the placebo  $p$  (the same signal shifted in time) was 0.72 and four of five gates failed.
- *H2, cost channel: consistent with the data, not established* **POST-HOC**. The hiring score has a 12-month IC of  $+0.096$  ( $t = 2.43$ ;  $+0.085$  on as-known data) and HQ12 is positive in all five reporting windows, but it lost money in the first half of the test and its deflated Sharpe is 0.05. A forward test is live.
- *H3, tightness: untestable as designed* **POST-HOC**.  $V/U$  never exceeded 1 in the research sample, so 133 of 142 test months fell in one tightness bin.

*Study 1: the one clean test.* The upper block of Table 6 is the result the paper stands on. Five H1 strategies were fixed on research data and run once on the test months; all five lost money, and trading costs are not the explanation.

TABLE 6. Net Sharpe ratio of every strategy (and net annual return in the test): research, the test, and the brief's three windows. **CONFIRMATORY** (Study 1) / **POST-HOC** (Study 2)

| Strategy                                                                                             | Research | Test  | As known | Net return | Full  | Post-2010 | Last 18m |
|------------------------------------------------------------------------------------------------------|----------|-------|----------|------------|-------|-----------|----------|
| <i>Study 1, preregistered: the one-time test (on as-known data)</i>                                  |          |       |          |            |       |           |          |
| A0 fixed rule                                                                                        | -0.44    | -0.36 |          | -3.5%      | -0.48 | -0.41     | -1.56    |
| A1 ridge                                                                                             | -0.54    | -0.51 |          | -4.0%      | -0.52 | -0.66     | +0.73    |
| A1-T ridge $\times$ tightness                                                                        | -0.55    | -0.59 |          | -4.9%      | -0.58 | -0.73     | -0.37    |
| A3 agent features, independent (primary)                                                             | -0.13    | -0.58 |          | -4.6%      | -0.44 | -0.62     | +0.85    |
| A4 agent features, communicating                                                                     | -0.18    | -0.79 |          | -6.1%      | -0.56 | -0.74     | +1.23    |
| <i>Study 2, post hoc (fixed-lag data; "As known" is the test Sharpe on as-known data, Study 2.3)</i> |          |       |          |            |       |           |          |
| W wage growth (declared headline)                                                                    | +0.09    | +0.16 | +0.13    | +1.7%      | +0.15 | +0.12     | -1.71    |
| SC signed composite                                                                                  | -0.06    | -0.15 | +0.24    | -1.3%      | -0.11 | -0.16     | -1.34    |
| W+MOM wage growth + momentum                                                                         | +0.27    | +0.35 | +0.46    | +3.1%      | +0.32 | +0.24     | -0.78    |
| HQ12 hires + quits reversal, 12 months                                                               | +0.50    | +0.29 | +0.20    | +2.6%      | +0.35 | +0.30     | +0.29    |

Annualised, net of 10 bp and 2 bp costs. Research: 88 common months for Study 1, May 2004 to September 2014 for Study 2. Test: November 2014 to August 2026. Full sample: from June 2007 for Study 1, from May 2004 for Study 2. Last 18 months: March 2025 to August 2026. A3 in the test: volatility 7.8%, maximum drawdown 41%, trading costs 1.1% a year against a gross loss of 3.5%.

Five things went wrong, all diagnosed after the fact (figures in Appendix B). *Signs*: layoffs carry a positive IC and hours a negative one, the opposite of the signs the fixed rule imposed (Table 1), and wage growth, dropped during the build, is the only signal whose IC exceeds 1.5 standard errors in both periods. *Horizon*: at one month no hiring signal predicts returns; at twelve months hires ( $-0.062$ ,  $t = -2.1$ ) and quits ( $-0.065$ ,  $t = -1.7$ ) carry the H2 sign. *Moderator*: tightness terciles were set on an expanding window, so 133 of 142 test months fell in the high tercile and H3 was never

tested. *Agent features*: the two features selected for A3 were set to zero outside the low-tightness tercile and so active in 3 of 142 test months, and every selected feature in both agent studies failed the test criterion. *Cap*: the gross cap bound in 87% of months, so realised volatility was 7.8% against a 10% target.

### 3.1 Style and risk exposures: partly a quality and growth tilt

We regress each book's net return on the Fama–French five factors, momentum and an industry-momentum factor (Table 7; every loading in Figure 12, Appendix C). The H1 books are quality and growth tilts rather than bets on labor news: industries adding workers and hours are the profitable, growing ones, so a long-hiring book buys profitability and sells value. The wage-growth book tilts the same way and adds momentum. HQ12 loads the other way, against profitability, which is what a book short the fastest hirers should do. None has a significant alpha.

TABLE 7. Factor regressions in the test window: alpha and the loadings with  $|t| > 2$ . CONFIRMATORY (A3) / POST-HOC (W, HQ12)

| Book                           | Alpha per month ( $t$ ) | Loadings with $ t  > 2$                                       |
|--------------------------------|-------------------------|---------------------------------------------------------------|
| A3 agent features, independent | −0.38% (−1.86)          | profitability RMW +0.19 ( $t = 2.2$ ); value HML −0.15 (−2.0) |
| W wage growth                  | $t = 0.29$              | value HML −0.43 (−4.4); momentum +0.22 (3.3)                  |
| HQ12 hires + quits reversal    | +0.26% (1.48)           | profitability RMW −0.36 (−4.45); size SMB −0.17 (−2.24)       |

### 3.2 Performance over time: every preregistered book lost

The last three columns of Table 6 are the windows the brief asks for. Every Study 1 book lost money over the full sample and after 2010. A3's last 18 months look good (+0.85) but rest on an IC  $t$  of 0.44. The post-hoc books show the opposite pattern: wage growth lost heavily in the last 18 months (−1.71), while HQ12 is positive in all five windows (cumulative returns in Figure 4, Appendix B).

### 3.3 Robustness and limits: why H1 failed, what survives

*Study 1 is negative under every check.* In Table 8 no data version, cost level, sub-period, dropped group, cap or weighting scheme turns A3 positive; its bootstrap interval excludes zero on the negative side, and its score has no IC at any horizon up to 24 months (Figure 3; −0.013 at 12 months).

TABLE 8. Study 1 robustness, test window. Upper block: from Study 1's saved results. Lower block: diagnostics added in Study 2.3. CONFIRMATORY / POST-HOC (lower block)

| Check                                                                  | A3 agent features (primary)           | A0 fixed rule                  |
|------------------------------------------------------------------------|---------------------------------------|--------------------------------|
| Sharpe by data version: as known / first release / revised / fixed lag | −0.58 / −0.52 / −0.56 / −0.53         | −0.36 / −0.16 / −0.26 / −0.27  |
| One-month IC ( $t$ ); factor alpha per month ( $t$ )                   | −0.021 (−1.08); −0.38% (−1.86)        | −0.020 (−0.71); −0.25% (−0.92) |
| Placebo $p$ ; Holm-adjusted $p$ of every registered comparison         | 0.72; 1.00                            | n/a; 1.00                      |
| Sharpe: first half / second half / last 18 months                      | −0.72 / −0.49 / +0.85                 | −0.55 / −0.20 / −1.56          |
| Sharpe at 0 / 10 / 25 bp costs; drop-one-group range                   | −0.45 / −0.58 / −0.79; −0.70 to −0.24 | not saved                      |
| Block-bootstrap 90% interval of the Sharpe (12-month blocks)           | [−0.99, −0.16]                        | [−0.91, +0.20]                 |
| Sharpe with the cap at 2 / 3 / none; rank-weighted book                | −0.58 / −0.61 / −0.62; −0.49          | not run                        |

*What survives: the 12-month cost channel.* We froze four post-hoc variants in a hashed file, ran each once and report all four (Table 6; statistics in Table 28, Appendix C). None meets the reading rule declared with them, which asks among other things for IC  $t \geq 2$ , alpha  $t \geq 2$  and a DSR above 0.95. The declared headline W has the right IC (+0.049,  $t = 2.04$ ) but no alpha ( $t = 0.29$ ); W+MOM reaches +0.35 because momentum supplies the return; SC is negative, and its sign flips on as-known data because three of its learned signs change with the data version. On as-known data W+MOM ranks first and HQ12 third (Table 6); we follow HQ12 because it is the H2 book, chosen on the hypothesis and not on its Sharpe ratio. HQ12 is the H2 book. Its sign is the one the investment channel predicts, but its 12-month horizon was chosen from a profile that included the test period: on the research years alone its 12-month IC is +0.047 ( $t = 0.90$ ), so the research sample would not have selected it. The as-known and fixed-lag scores pick the same top three groups in only 39% of months, so real-time holdings would have differed. Table 9 runs it through the pre-declared battery on both data versions: it keeps its sign and most of its size as known, and fails the same check in both, the first test half.

TABLE 9. HQ12 robustness battery, test window unless stated, on revised data with fixed lags (Study 2.2) and on data as known at each decision (Study 2.3). The last column says whether the pre-declared condition is met on both data versions.

| Check                                               | Fixed lag                      | As known                       | Met on both? |
|-----------------------------------------------------|--------------------------------|--------------------------------|--------------|
| Net Sharpe; 12-month IC ( $t$ )                     | +0.29; +0.096 (2.43)           | +0.20; +0.085 (2.26)           | yes          |
| Components held 12 months: hires alone, quits alone | +0.19, +0.16                   | +0.19, +0.17                   | yes          |
| Holding 9 and 18 months                             | +0.14, +0.30                   | +0.16, +0.11                   | yes          |
| Both test halves positive                           | −0.21, +0.64                   | −0.23, +0.49                   | no           |
| Break-even cost                                     | 68 bp                          | 50 bp                          | yes          |
| Drop one group: lowest and highest Sharpe           | +0.18, +0.48                   | +0.05, +0.36                   | yes          |
| Bootstrap 90% interval: test; full sample           | [−0.07, +0.65]; [+0.05, +0.64] | [−0.14, +0.57]; [+0.02, +0.60] | full only    |
| Alpha $t$ (12 lags): test; full sample              | 1.48; 1.92                     | 0.92; 1.81                     | no           |
| Deflated Sharpe (looks counted)                     | 0.05 (132)                     | 0.02 (162)                     | no           |

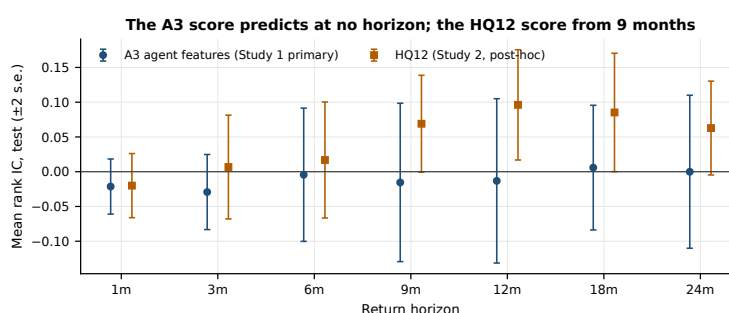


FIGURE 3. Horizon analysis, test window: the A3 score has no IC at any horizon; the HQ12 score has none at one month and a positive IC from 9 to 24 months.

*Looks, forward test and limits.* Rank weights, the cap, breadth and capacity are in Section 2.3 and do not change the picture. We took 162 looks in all. With 142 test months only a Sharpe near 0.6 reaches  $t \approx 2$ , so a null does not rule out a smaller effect, and the best post-hoc test Sharpe among all looks, W+MOM's +0.46 on as-known data, corresponds to  $t \approx 1.6$ . Four paper books (W, SC, W+MOM, HQ12) were frozen on 3 October 2026 for the 30 September decision and are held to 31 December with no capital behind them (Appendix D); one quarter has little power, and no position of any size is recommended. Stock-level baskets by NAICS code from CRSP are the study we would run next.

**TABLE 10.** Claude as research assistant: prompt, summarised output, adaptation and critical evaluation (prompts in Section 2.3, detail and transcripts in Appendix E). CONFIRMATORY (P1–P4) / POST-HOC (P5–P6)

| Interaction and prompt                                                                                                                                                                                                                                                                                                                                                                             | Output: what was right                                                                                                   | Output: what was wrong                                                                                                                   | Adaptation                                                                                           | Our critical evaluation                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| P1 data and code audit (Sonnet): find mapping, look-ahead and code bugs in the manifest, crosswalk and as-of code                                                                                                                                                                                                                                                                                  | Flagged every real scope gap: transport without utilities, Telcm for Information, groups 9 and 10 sharing one CES series | Called design choices “confirmed errors”; 5 of 25 findings wrong (the “vanished” group-10 series had 584 shared rows); no look-ahead bug | All 25 tested; one metadata fix to the crosswalk; no number or timing rule changed                   | [TODO: critique: Alex]                                                                                                        |
| P2 signal agents (Sonnet; 3 independent, 3 communicating, 3 seeds): blind to names and dates, propose six features each in a 14-operation grammar                                                                                                                                                                                                                                                  | 98 of 108 proposals valid; blinding audit of all 199 calls passed; best research IC 0.08 independent, 0.05 communicating | Both independent-arm picks switch on only in the low-tightness tercile; 11 of 18 migration reports broke the rules, 10 over 120 words    | Invalid proposals used their slot; selection mechanical, two per arm, seed 1 only; no engineer edits |                                                                                                                               |
| P3 judge (Sonnet; Opus in Study 1b): does a later proposal use or test a claim from an earlier report?                                                                                                                                                                                                                                                                                             | Study 1b: agreed with the human on 10 of 17 pairs; caught all 3 human “yes” cases                                        | Study 1: 28 of 36 answers not clean JSON, 8 usable; Study 1b: all 7 disagreements false “yes”                                            | Undefined labels kept as missing; no repair parsing or extra calls; uptake reported as descriptive   | IC shown by tightness tercile pushed agents to regime gates; both A3 picks were off for most of the test<br>see the row below |
| P3, our critical evaluation: All 7 disagreements were false positives, not a random mix – a judge that scores on topic/vocabulary overlap will drift toward “yes” wherever reports share wording. With only 17 audited pairs we can’t rule out noise, but the one-sided error pattern points to a concrete fix: require the judge to quote the exact borrowed claim, not just flag a shared topic. |                                                                                                                          |                                                                                                                                          |                                                                                                      |                                                                                                                               |
| P4 red team (Sonnet): from anonymous research tables, ten reasons the results could be spurious, each with a test                                                                                                                                                                                                                                                                                  | Read the tables correctly: no IC $t$ above 1.3, negative Sharpes; 6 of 10 concerns matched registered tests              | Stated a permutation $p$ it never computed; proposed thresholds after seeing results (95th percentile, DSR 0.95, rolling 24 months)      | Registered diagnostics kept; the 4 new tests and every new threshold rejected as post hoc            | [TODO: critique: Romain]                                                                                                      |
| P5 post-mortem review (Opus, after the test): why did Study 1 lose money?                                                                                                                                                                                                                                                                                                                          | Agent features active in about 3 of 142 test months, which fits the low- $T$ gate: $T$ stayed high after 2014            | Its fixes (wage growth back, two A0 signs flipped) read the test results; its “stable” wage signal: $-1.71$ over the last 18 months      | Study 2.1: four variants frozen before running, all reported, W the headline; labelled post hoc      | [TODO: critique: Piero]                                                                                                       |
| P6 iteration (Opus, Fable): rebuild the analysis, run Study 2, draft this report                                                                                                                                                                                                                                                                                                                   | Reproduced Study 1’s returns to $10^{-16}$ ; froze every spec before running; stopped at a failed gate                   | Assumed August JOLTS would not be out by 29 September; it was, and its own check stopped the run                                         | Forward-test amendment 001, disclosed (Appendix D); run logs keep results reproducible               | [TODO: critique: Alex]                                                                                                        |

### 3.4 Claude evaluation: useful auditor, unreliable judge

Claude was the AI research assistant in every role, which this term’s version of the course allows; the switch from the ChatGPT template is logged as preregistration amendments 001 and 002 of Study 1. The models are taken from the call logs: Sonnet 5.5 at medium effort for the 199 calls of Study 1, Opus 5.5 at extra-high effort for the 294 calls of Study 1b, an exploratory rerun of the agent experiment with a stronger model (Appendix B). Alex used a general-purpose coding assistant (“6 astra”) for engineering support; it took no part in the study’s research, agent or evaluation roles. Table 10 documents each interaction: the prompt, a summary of the output, our critical evaluation and what we changed.

*The agent experiment, tested further.* POST-HOC Sharing did not help: in both agent studies the gap in best research IC between the communicating and the independent arm is within the spread across seeds (permutation  $p = 0.30$  and  $0.10$ , the smallest value 20 relabelings allow), and tokens per call are similar across arms. We then tested the two fixes our critiques propose. Requiring the judge to quote the claim it relies on made the Sonnet judge’s answers usable (8 of 8, 7 in agreement with the human, against 2 usable before) but did not cure the Opus judge: only 7 of its 17 answers were in the required format, and counting the rest it still said yes too often (8 false yes in 17, against 7). Removing the tightness-tercile split from the agents’ feedback did not remove regime gates: 4 of 48 valid proposals used one with the split and 3 of 50 without, on research data only. So the feedback explanation in

Table 10 is not confirmed, although the rerun also produced fewer gates than the original run (15% to 29% per seed).

*Reflection.* Claude was good at fast, wide first-pass audits (P1), at structured scepticism (P4) and at following a blind protocol (98 of 108 valid proposals; blinding audit passed). It was bad at calibration (design choices became “confirmed errors”), at holding rules fixed (new thresholds after seeing results), at counting and formats, and at judging provenance (7 false yes in 17). Claude is a useful auditor and idea generator when a human checks every claim against the code and data, and when it cannot change the rules after seeing results.

## References

---

- David H. Bailey and Marcos López de Prado. The deflated sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5):94–107, 2014.
- Frederico Belo, Xiaoji Lin, and Santiago Bazdresch. Labor hiring, investment, and stock return predictability in the cross section. *Journal of Political Economy*, 122(1):129–177, 2014.
- Andres Donangelo, François Gourio, Matthias Kehrig, and Miguel Palacios. The cross-section of labor leverage and equity returns. *Journal of Financial Economics*, 132(2):497–518, 2019.
- Harrison Hong, Walter Torous, and Rossen Valkanov. Do industries lead stock markets? *Journal of Financial Economics*, 83(2):367–396, 2007.
- Lars-Alexander Kuehn, Mikhail Simutin, and Jessie Jiaxu Wang. A labor capital asset pricing model. *Journal of Finance*, 72(5):2295–2332, 2017.
- Olivier Ledoit and Michael Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004.
- Tobias J. Moskowitz and Mark Grinblatt. Do industries explain momentum? *Journal of Finance*, 54(4):1249–1290, 1999.
- Carlo Zarattini and Gary Antonacci. A century of profitable industry trends. SSRN Working Paper 4857230, 2024.

## A Names used in this report

The repository and the result files were renamed on 4 October 2026. Result files frozen before that date keep the old codes inside, so they stay byte-identical.

TABLE 11. Name mapping.

| Old name                       | Name in this report                                                                                                                |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| frozen, sealed or Sonnet study | Study 1: preregistered (Alex), folder <code>study1_preregistered/</code>                                                           |
| Opus follow-up                 | Study 1b: exploratory follow-up (Opus)                                                                                             |
| v2, v3, v4 runs                | Study 2: post-hoc iteration; 2.1 variants (was v2), 2.2 robustness of HQ12 (was v3), 2.3 additional robustness (was v4)            |
| A0, A1, A1-T, A3, A4           | A0 fixed rule; A1 ridge; A1-T ridge $\times$ tightness; A3 agent features, independent (primary); A4 agent features, communicating |
| V2a, V2b, V2c, V2d             | W wage growth (declared headline); SC signed composite; W+MOM wage growth + momentum; HQ12 hires + quits reversal, 12-month hold   |
| P1 to P6                       | P1 data and code audit; P2 signal agents; P3 judge; P4 red team; P5 post-mortem review; P6 iteration                               |
| islands, sealed test, F1–F5    | communicating arm; one-time test; openings, hires, quits, layoffs, hours (Table 1)                                                 |

## B Study 1 in full: preregistered (Alex)

Everything in this section labelled confirmatory is read from Study 1’s saved files and is unchanged: `results.json` and the ledgers next to it, under `study1_preregistered/outputs/follow_the_workers/outputs/sealed/initial/`. The diagnostics labelled post-hoc were computed afterwards from the same saved files and public data.

### Design

Thirteen industry groups (Appendix H); signals F1–F5 as known at each decision date; five strategies (Table 3); research May 2004 to September 2014, test November 2014 to August 2026; top three against bottom three, three monthly tranches, beta hedge, 10% volatility target, gross cap 2, 10 bp and 2 bp costs. Five gates decide the verdict for the primary: G1 timing (pass), G2 prediction (IC  $t \geq 2$ ; fail), G3 incremental alpha ( $t \geq 2$ ; fail), G4 robustness (positive IC in both halves, placebo  $p < 0.05$ , no industry above half the profit; fail), G5 implementable (Sharpe  $> 0$  at 25 bp, capacity documented; fail). The freeze record was checked against the files (Table 17). The score-to-weight mapping, with  $s_{m,g}$  the score of group  $g$  known at the end of month  $m - 1$ :

$$\tau_{m,g} = \begin{cases} +\frac{1}{3} & g \in \text{top 3 of } s_{m,\cdot} \\ -\frac{1}{3} & g \in \text{bottom 3 of } s_{m,\cdot} \\ 0 & \text{otherwise} \end{cases} \quad b_m = \frac{1}{H} \sum_{j=0}^{H-1} \tau_{m-j}, \quad H = 3, \quad (3)$$

$$h_m = - \sum_g b_{m,g} \hat{\beta}_{m,g}, \quad x_m = (b_{m,1}, \dots, b_{m,13}, h_m), \quad (4)$$

$$k_m = \min \left( \frac{0.10}{\sqrt{12 x_m^\top \hat{\Sigma}_m x_m}}, \frac{2}{\|x_m\|_1} \right), \quad w_m = k_m x_m, \quad (5)$$

$$r_m^{\text{net}} = \sum_g w_{m,g} r_{m,g}^e + w_{m,M} \text{MKT}_m - c \sum_g |w_{m,g} - \tilde{w}_{m,g}| - c_h |w_{m,M} - \tilde{w}_{m,M}|. \quad (6)$$

where  $s_{m,g}$  is the score of group  $g$  known at the end of month  $m - 1$ ,  $\hat{\beta}_{m,g}$  and the Ledoit–Wolf covariance  $\hat{\Sigma}_m$  (13 groups plus the market) use the 60 months ending in  $m - 2$  (at least 36),  $r_{m,g}^e$  is the group’s excess return,  $\tilde{w}$  are the previous weights after drift,  $c = 10$  bp and  $c_h = 2$  bp.

TABLE 12. Study 1: all five strategies in the test window.

CONFIRMATORY

| strategy                | Sharpe | Net p.a. | Vol. | Max DD | Turnover | Costs p.a. |
|-------------------------|--------|----------|------|--------|----------|------------|
| A0 fixed rule           | −0.36  | −3.5%    | 9.7% | −40.3% | 0.83     | 1.01%      |
| A1 ridge                | −0.51  | −4.0%    | 7.9% | −37.7% | 0.91     | 1.11%      |
| A3 primary              | −0.58  | −4.6%    | 7.8% | −41.1% | 0.91     | 1.11%      |
| A4 communicating agents | −0.79  | −6.1%    | 7.8% | −53.3% | 1.11     | 1.35%      |
| Industry momentum 12-1  | +0.03  | +0.3%    | 9.4% | −20.8% | 0.53     | 0.65%      |

TABLE 13. Study 1 robustness, full table: the four data versions, costs, sub-periods, drop-one-group, gates, and the diagnostics added in Study 2.3.

CONFIRMATORY / POST-HOC (last four rows)

| Check (test window, as-known data unless stated)                              | A3 agent features (primary)                                       | A0 fixed rule         |
|-------------------------------------------------------------------------------|-------------------------------------------------------------------|-----------------------|
| Test Sharpe, data as known                                                    | −0.58                                                             | −0.36                 |
| Test Sharpe, data first release                                               | −0.52                                                             | −0.16                 |
| Test Sharpe, data revised                                                     | −0.56                                                             | −0.26                 |
| Test Sharpe, data fixed lag                                                   | −0.53                                                             | −0.27                 |
| Net return p.a.                                                               | −4.6%                                                             | −3.5%                 |
| IC, one month (t)                                                             | −0.021 (−1.08)                                                    | −0.020 (−0.71)        |
| Factor alpha per month (t)                                                    | −0.38% (−1.86)                                                    | −0.25% (−0.92)        |
| Placebo p (circular shift)                                                    | 0.72                                                              | n/a (primary only)    |
| Holm-adjusted p, IC and return                                                | 1.00, 1.00 (A3 vs A1)                                             | 1.00, 1.00 (A1 vs A0) |
| Deflated Sharpe, research (41 trials)                                         | 0.003                                                             | 0.000                 |
| Sharpe at 0 / 10 / 25 bp                                                      | −0.45 / −0.58 / −0.79                                             | n/a / −0.36 / n/a     |
| Sharpe, first test half                                                       | −0.72                                                             | −0.55                 |
| Sharpe, second test half                                                      | −0.49                                                             | −0.20                 |
| Sharpe, excluding Mar-Dec 2020                                                | −0.51                                                             | −0.18                 |
| Sharpe, last 18 months                                                        | +0.85                                                             | −1.56                 |
| Drop one group, Sharpe range                                                  | −0.70 (Wholesale) to −0.24 (Durable mfg)                          | n/a (primary only)    |
| Gates G1 timing / G2 prediction / G3 alpha / G4 robustness / G5 implementable | pass / fail / fail / fail / fail                                  | n/a (primary only)    |
| Verdict                                                                       | Do not implement                                                  | not primary           |
| Block-bootstrap 90% interval, test Sharpe                                     | [−0.99, −0.16]                                                    | [−0.91, +0.20]        |
| A3 score IC at 1 / 6 / 12 / 24 months (t)                                     | −0.021 (−1.08) / −0.004 (−0.09) / −0.013 (−0.22) / −0.000 (−0.00) | n/a                   |
| A3 Sharpe with cap 2 / cap 3 / no cap                                         | −0.58 / −0.61 / −0.62                                             | n/a                   |
| A3 Sharpe, rank-weighted book                                                 | −0.49                                                             | n/a                   |

TABLE 14. A3 by window, from the saved results.

CONFIRMATORY

| window                               | Months | Sharpe | Net return p.a. | Mean IC (t)    |
|--------------------------------------|--------|--------|-----------------|----------------|
| Full test, Nov 2014-Aug 2026         | 142    | −0.58  | −4.6%           | −0.021 (−1.08) |
| First half                           | 71     | −0.72  | −4.9%           | −0.028 (−0.90) |
| Second half                          | 71     | −0.49  | −4.3%           | −0.015 (−0.59) |
| Excluding Mar-Dec 2020               | 132    | −0.51  | −4.1%           | −0.012 (−0.56) |
| Last 18 months (Mar 2025-Aug 2026)   | 18     | +0.85  | +6.7%           | +0.033 (+0.44) |
| Course window "full" (from Jun 2007) | 231    | −0.44  | −3.2%           | −0.009 (−0.49) |
| Course window "post-2010"            | 200    | −0.62  | −4.4%           | −0.019 (−1.02) |

TABLE 15. Risk overlay of every Study 1 and Study 1b strategy: how often the cap bound, ex-ante and realised volatility.

CONFIRMATORY

| Study, strategy | Cap binds | Ex-ante vol | Realised vol | Gross leverage | Hedge |
|-----------------|-----------|-------------|--------------|----------------|-------|
| Study 1 A0      | 80%       | 8.0%        | 9.7%         | 1.96           | 0.15  |
| Study 1 A1      | 88%       | 7.5%        | 7.9%         | 1.98           | 0.13  |
| Study 1 A1-T    | 82%       | 7.5%        | 8.2%         | 1.97           | 0.12  |
| Study 1 A3      | 87%       | 7.5%        | 7.8%         | 1.98           | 0.13  |
| Study 1 A4      | 91%       | 7.3%        | 7.8%         | 1.99           | 0.13  |
| Study 1b A3     | 92%       | 7.3%        | 7.5%         | 1.99           | 0.15  |
| Study 1b A4     | 93%       | 7.3%        | 8.4%         | 1.99           | 0.11  |

TABLE 16. Factor loadings of the Study 1 books (FF5, momentum, industry momentum).

CONFIRMATORY

| Factor            | A0 fixed rule  | A1 ridge       | A3 primary     |
|-------------------|----------------|----------------|----------------|
| Alpha (per month) | −0.25% (−0.92) | −0.35% (−1.72) | −0.38% (−1.86) |
| Market            | −0.01 (−0.31)  | −0.01 (−0.20)  | −0.03 (−0.43)  |
| SMB               | +0.02 (+0.21)  | +0.12 (+1.01)  | +0.10 (+0.89)  |
| HML               | +0.09 (+0.60)  | −0.15 (−1.97)  | −0.15 (−2.01)  |
| RMW               | +0.00 (+0.05)  | +0.21 (+2.33)  | +0.19 (+2.16)  |
| CMA               | −0.04 (−0.21)  | +0.05 (+0.47)  | +0.03 (+0.26)  |
| Momentum          | −0.06 (−0.72)  | +0.01 (+0.15)  | −0.01 (−0.17)  |
| Industry momentum | +0.22 (+1.93)  | −0.02 (−0.21)  | −0.01 (−0.12)  |

TABLE 17. Freeze record, checked against the files in the repository.

CONFIRMATORY

| check                                          | ok   | value                             |
|------------------------------------------------|------|-----------------------------------|
| SHA-256 of FREEZE.json equals evaluation       | True | 60ea05294ffc856e                  |
| Freeze recorded (UTC)                          | True | 2026-10-02T06:00:09.537758+00:00  |
| Test opened (UTC)                              | True | 2026-10-02T06:00:09.578162+00:00  |
| Test opened after the freeze                   | True | 0.040 s later                     |
| Files in the freeze inventory                  | True | 525                               |
| of which present in the repository             | True | 338                               |
| present files whose hash matches the inventory | True | 337                               |
| present files that differ                      | True | outputs/human_migration_review.md |
| First test decision in the freeze record       | True | 2014-10-31                        |

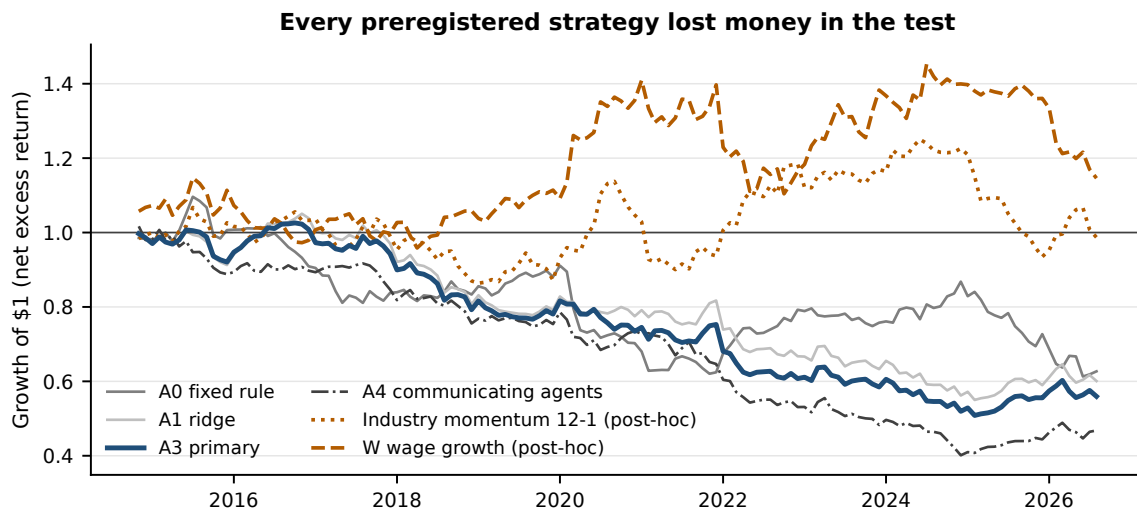


FIGURE 4. Cumulative net returns in the test: every Study 1 strategy lost money; the post-hoc wage-growth book gained until 2024 and gave much of it back.

CONFIRMATORY (A0–A4) / POST-HOC (momentum, W)

## Diagnostics of the failure

TABLE 18. One-month rank IC of each signal by period.

POST-HOC

| Signal                 | Research 2004–14 | Test 2014–26   | Full 2004–26   |
|------------------------|------------------|----------------|----------------|
| F1 openings            | +0.001 (+0.02)   | −0.011 (−0.44) | −0.005 (−0.30) |
| F2 hires               | +0.012 (+0.48)   | +0.013 (+0.48) | +0.013 (+0.71) |
| F3 quits               | +0.014 (+0.49)   | +0.002 (+0.11) | +0.009 (+0.50) |
| F4 layoffs             | +0.033 (+1.07)   | +0.026 (+0.93) | +0.028 (+1.38) |
| F5 hours               | −0.012 (−0.55)   | −0.007 (−0.34) | −0.010 (−0.67) |
| W wage growth          | +0.029 (+1.54)   | +0.049 (+2.04) | +0.039 (+2.50) |
| A0 composite (prereg.) | −0.023 (−0.88)   | −0.006 (−0.21) | −0.013 (−0.69) |
| Industry momentum 12-1 | +0.036 (+1.25)   | +0.003 (+0.11) | +0.020 (+0.99) |

TABLE 19. Rank IC by horizon of every signal, May 2004–August 2026 ( $t$  in brackets).

POST-HOC

| Signal                 | 1m             | 3m             | 6m             | 12m            |
|------------------------|----------------|----------------|----------------|----------------|
| F1 openings            | −0.005 (−0.31) | +0.007 (+0.23) | −0.011 (−0.28) | −0.039 (−0.92) |
| F2 hires               | +0.013 (+0.70) | +0.002 (+0.07) | +0.002 (+0.08) | −0.062 (−2.08) |
| F3 quits               | +0.009 (+0.50) | −0.013 (−0.55) | −0.022 (−0.78) | −0.065 (−1.73) |
| F4 layoffs             | +0.028 (+1.37) | +0.005 (+0.17) | +0.000 (+0.01) | −0.036 (−1.28) |
| F5 hours               | −0.010 (−0.67) | −0.027 (−1.24) | −0.049 (−1.69) | −0.057 (−1.86) |
| W wage growth          | +0.039 (+2.51) | +0.037 (+1.77) | +0.037 (+1.28) | +0.026 (+0.63) |
| A0 composite (prereg.) | −0.013 (−0.69) | −0.002 (−0.09) | −0.021 (−0.61) | −0.053 (−1.58) |
| Industry momentum 12-1 | +0.020 (+0.99) | +0.025 (+0.80) | +0.048 (+1.24) | +0.029 (+0.63) |

TABLE 20. Tightness regimes: months per bin under the expanding-tercile rule and under a 12-month change rule.

POST-HOC

| Months in regime | Research 2004–14 | Test 2014–26 |
|------------------|------------------|--------------|
| months           | 125              | 142          |
| expanding low    | 49               | 3            |
| expanding middle | 26               | 6            |
| expanding high   | 50               | 133          |
| T rising (12m)   | 93               | 88           |
| T falling (12m)  | 32               | 54           |
| V/U > 1          | 0                | 75           |
| T min            | −1.88            | −1.61        |
| T max            | −0.30            | +0.71        |

TABLE 21. Signal IC when tightness rose or fell over the previous 12 months.

POST-HOC

| Signal                 | T rising (182 m) | T falling (86 m) | Difference |
|------------------------|------------------|------------------|------------|
| F1 openings            | +0.001 (+0.05)   | −0.019 (−0.47)   | +0.019     |
| F2 hires               | +0.032 (+1.51)   | −0.026 (−0.74)   | +0.059     |
| F3 quits               | +0.024 (+1.32)   | −0.024 (−0.76)   | +0.048     |
| F4 layoffs             | +0.013 (+0.53)   | +0.060 (+1.69)   | −0.046     |
| F5 hours               | −0.002 (−0.13)   | −0.026 (−1.00)   | +0.024     |
| W wage growth          | +0.033 (+1.67)   | +0.053 (+2.04)   | −0.020     |
| A0 composite (prereg.) | +0.003 (+0.13)   | −0.045 (−1.28)   | +0.048     |
| Industry momentum 12-1 | +0.036 (+1.68)   | −0.014 (−0.33)   | +0.050     |

TABLE 22. A3 profit and loss by industry in the test.

POST-HOC

| component             | Contribution (pp) | Months long | Months short |
|-----------------------|-------------------|-------------|--------------|
| Mining & logging      | +7.2              | 33%         | 36%          |
| Construction          | −19.2             | 36%         | 39%          |
| Durable mfg           | +1.8              | 31%         | 24%          |
| Nondurable mfg        | +5.4              | 32%         | 35%          |
| Wholesale             | +14.9             | 31%         | 42%          |
| Retail                | +0.2              | 39%         | 35%          |
| Transport & utilities | +1.4              | 47%         | 30%          |
| Information           | −5.8              | 30%         | 37%          |
| Finance & insurance   | −16.3             | 34%         | 45%          |
| Real estate           | −1.6              | 24%         | 45%          |
| Prof. & business svcs | −12.2             | 18%         | 47%          |
| Health care           | −26.2             | 41%         | 36%          |
| Accommodation & food  | +4.5              | 46%         | 22%          |
| Market hedge          | +4.6              |             |              |
| Trading costs         | −13.1             |             |              |

TABLE 23. Horizon profile of the A3 score against the HQ12 score, test window (IC,  $t$  in brackets). POST-HOC

| A3 agent features (Study 1 primary) | HQ12 (Study 2, post-hoc) |
|-------------------------------------|--------------------------|
| -0.021 (-1.08)                      | -0.020 (-0.87)           |
| -0.029 (-1.08)                      | +0.007 (+0.18)           |
| -0.004 (-0.09)                      | +0.017 (+0.40)           |
| -0.015 (-0.27)                      | +0.069 (+1.98)           |
| -0.013 (-0.22)                      | +0.096 (+2.43)           |
| +0.006 (+0.13)                      | +0.085 (+2.00)           |
| -0.000 (-0.00)                      | +0.063 (+1.86)           |

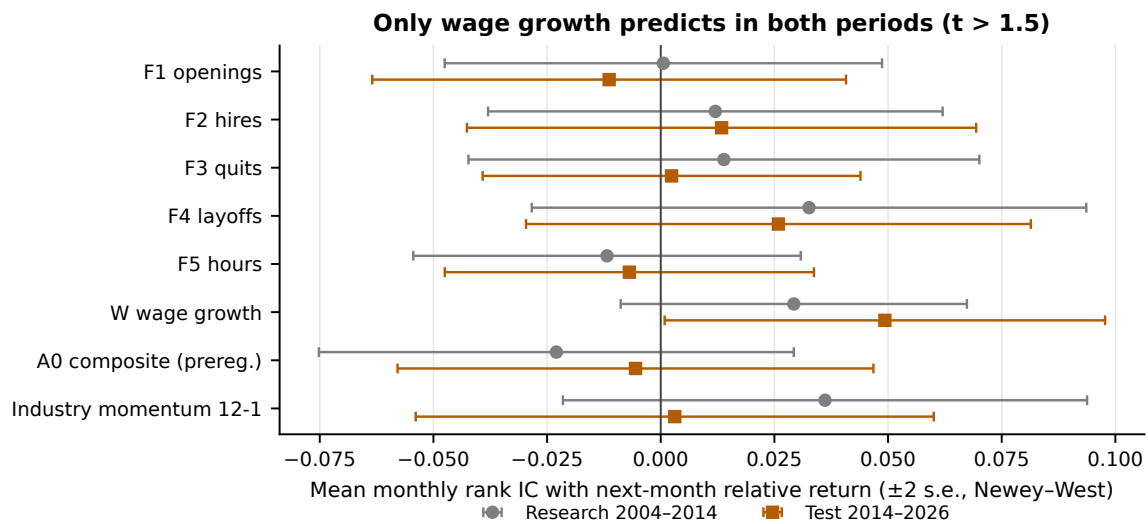
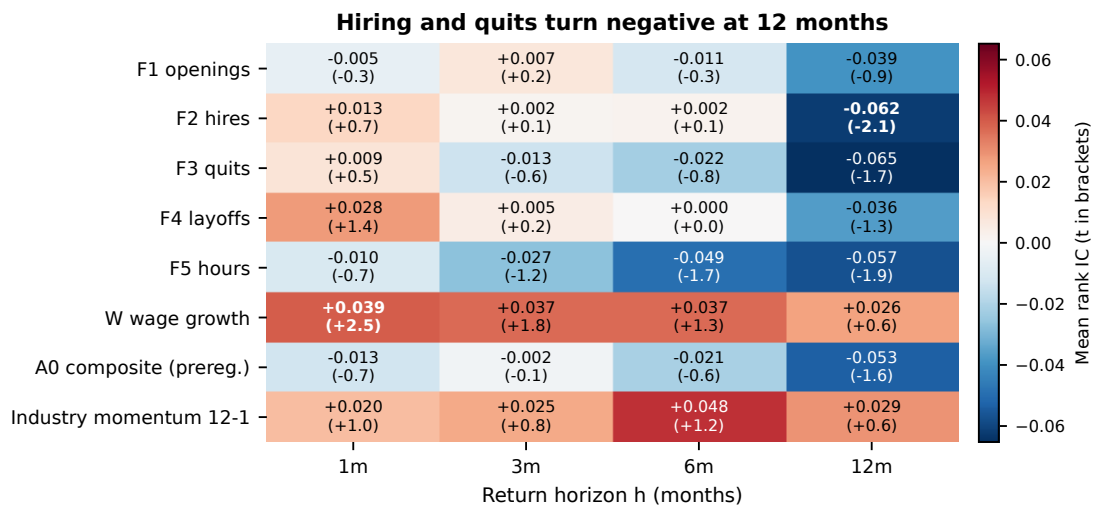


FIGURE 5. Only wage growth has an IC above 1.5 standard errors in both periods; layoffs and hours carry the opposite sign to the one the fixed rule imposed. POST-HOC

FIGURE 6. Hires and quits turn negative at 12 months ( $t = -2.1, -1.7$ ): the investment channel shows up slowly, not within a month. POST-HOC

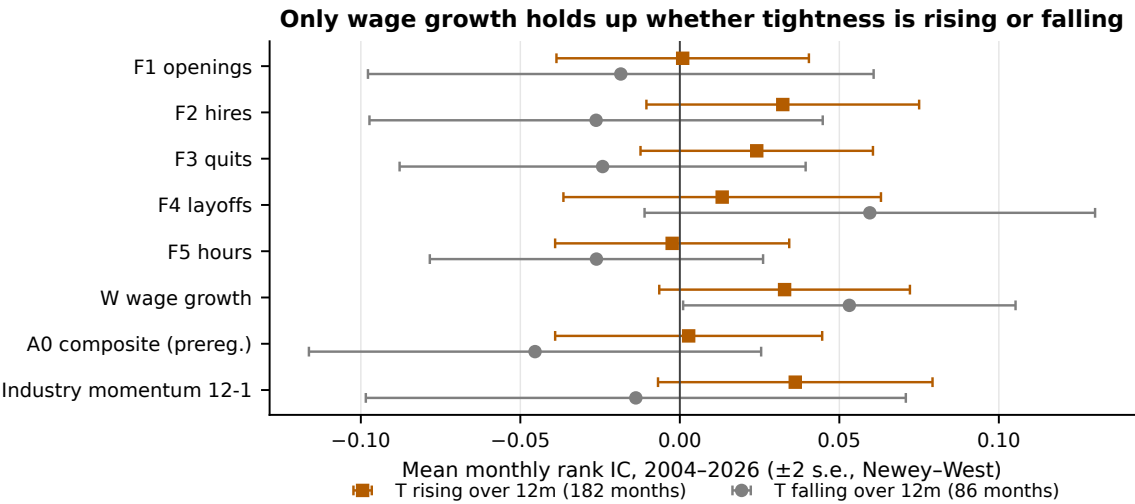


FIGURE 7. Single-signal IC in months when  $T$  rose versus fell over the previous 12 months: only wage growth keeps an IC above 1.5 standard errors in both states. POST-HOC

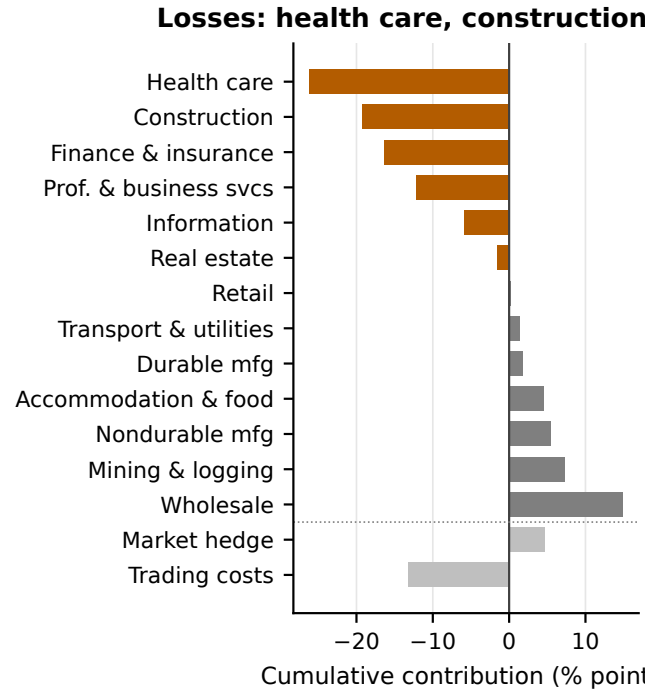


FIGURE 8. A3 lost most in health care, construction and finance. POST-HOC

## The agent experiment: design and results

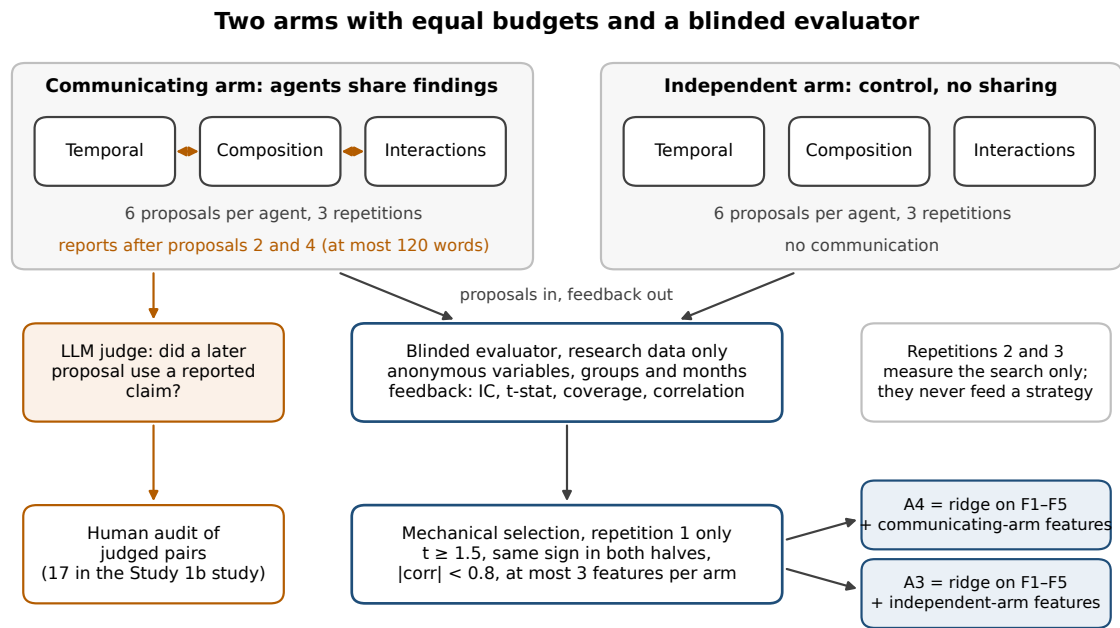


FIGURE 9. Two arms with equal budgets, a blinded evaluator, mechanical selection and a human-audited LLM judge.

TABLE 24. Agent experiment: design parameters and counts in both studies.

CONFIRMATORY

| study    | item                              | value                                                                              |
|----------|-----------------------------------|------------------------------------------------------------------------------------|
| Both     | Arms                              | communicating (communicating), independent (control)                               |
| Both     | Agents per arm                    | Temporal, Composition, Interactions                                                |
| Both     | Proposals per agent x repetitions | 6 x 3                                                                              |
| Both     | Migration reports                 | after proposals 2 and 4, at most 120 words                                         |
| Both     | Selection (repetition 1 only)     | $t \geq 1.5$ , same sign in both halves, $ \text{corr}  < 0.8$ , at most 3 per arm |
| Study 1  | Valid proposals                   | 98 / 108                                                                           |
| Study 1  | Migration reports admitted        | 7 / 18                                                                             |
| Study 1  | Defined LLM-judge uptake labels   | 8 / 36                                                                             |
| Study 1  | Human-audited judge pairs         | 8                                                                                  |
| Study 1b | Valid proposals                   | 106 / 108                                                                          |
| Study 1b | Migration reports admitted        | 15 / 18                                                                            |
| Study 1b | Defined LLM-judge uptake labels   | 84 / 88                                                                            |
| Study 1b | Human-audited judge pairs         | 17                                                                                 |
| Study 1b | Human-judge agreement             | 10 / 17 (human yes 3, judge yes 10)                                                |

TABLE 25. Agent features: how many passed the research filter and the test criterion.

POST-HOC

| Repetition-1 candidates                     | Study 1 | Study 1b |
|---------------------------------------------|---------|----------|
| valid candidates                            | 33.0    | 35.0     |
| regime-gated                                | 9.0     | 6.0      |
| test IC undefined ( $\leq 3$ active months) | 4.0     | 2.0      |
| corr(research IC, test IC)                  | 0.22    | 0.37     |
| passed research filter                      | 4.0     | 7.0      |
| selected                                    | 4.0     | 4.0      |
| selected and regime-gated                   | 2.0     | 2.0      |
| selected with positive test IC              | 0.0     | 2.0      |

TABLE 26. Selected features, decoded after the test: research and test IC, months active.

POST-HOC

| Study    | Arm           | Feature (decoded)                                                        | Research IC (n) | Test IC (n)  |
|----------|---------------|--------------------------------------------------------------------------|-----------------|--------------|
| Study 1  | independent   | regime(yoy(Earnings), "low")                                             | +0.065 (48)     | n/a (3)      |
| Study 1  | independent   | regime(spread(tsz(Layoffs, 60), tsz(Hires, 60)), "low")                  | +0.070 (43)     | n/a (3)      |
| Study 1  | communicating | spread(chg(Layoffs, 3), chg(Openings, 3))                                | +0.037 (123)    | −0.023 (142) |
| Study 1  | communicating | spread(chg(Layoffs, 6), chg(Openings, 6))                                | +0.052 (120)    | −0.004 (142) |
| Study 1b | independent   | mult(regime(tsz(Hires, "exp"), "high"),<br>csrank(tsz(Openings, "exp"))) | +0.088 (15)     | −0.028 (133) |
| Study 1b | communicating | tsz(lchg(lag(Earnings, 1), 12), 36)                                      | +0.064 (78)     | +0.022 (142) |
| Study 1b | communicating | regime(spread(chg(Quits, 3), chg(Hires, 3)), "low")                      | +0.125 (48)     | n/a (3)      |
| Study 1b | communicating | tsz(lchg(lag(Hours, 1), 12), 36)                                         | +0.057 (78)     | +0.011 (142) |

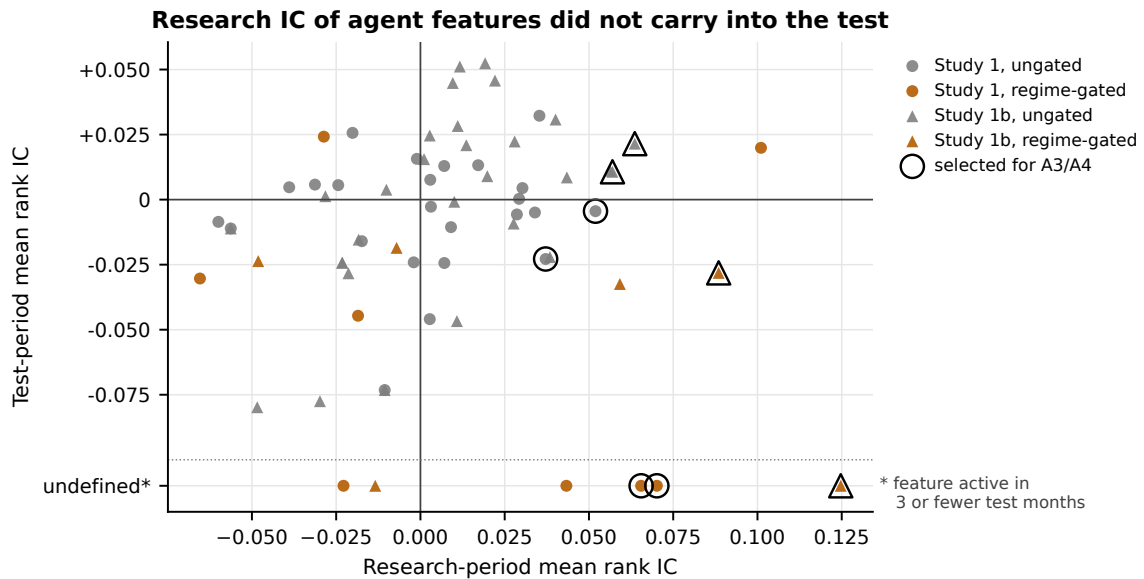


FIGURE 10. Research IC of agent features barely predicts their test IC; the selected regime-gated features were active in 3 test months.

POST-HOC

### Study 1b: exploratory follow-up with Opus

After the test had been opened, the agent experiment was rerun with Claude Opus 5.5 at extra-high effort, with fresh agents and the same evaluator, selection rule and portfolio. Because the test period was already known to the team, the run is exploratory. Its research-selected primary, A4 (agent features, communicating), earned a test Sharpe of +0.01 (research −0.40) with an IC of −0.002 ( $t = -0.10$ ), an alpha  $t$  of 0.05, a placebo  $p$  of 0.56 and a Sharpe of −0.19 at 25 bp; it failed gates G2 to G5 and had a Sharpe of −1.41 over the last 18 months. Its gross return came from one long position: durable manufacturing contributed +0.27 of cumulative gross return (sum of monthly contributions, in units of the book) in a group that is 60% semiconductors. Its A3 earned −0.46. The follow-up is evidence about agents, not about the strategy. A stronger model followed the protocol better and found no more signal: 106 of 108 valid proposals against 98, 15 of 18 admitted migration reports against 7, 84 of 88 usable judge labels against 8 of 36, yet no selected feature survived the test criterion in either study.

## C Study 2 in full: the post-hoc iteration

## Specifications, hashes and look counts

TABLE 27. Every Study 2 stage was written, hashed and committed before it ran. Times are UTC. **POST-HOC** / **FORWARD TEST**

| Specification (analysis/specs/) | SHA-256  | Frozen          | First run     | Looks counted in the deflated Sharpe                               |
|---------------------------------|----------|-----------------|---------------|--------------------------------------------------------------------|
| study2_1_variants.md            | ca658234 | 3 Oct, 09:00:13 | 09:01:55      | 112: about 108 diagnostic looks already taken, plus the 4 variants |
| forward_spec.md                 | 735a8ffa | 3 Oct, 09:03:56 | book 09:17:55 | none (forward test)                                                |
| forward_amendment_001.md        | c76cde86 | 3 Oct, 09:16:11 | book 09:17:55 | none                                                               |
| study2_2_hq12_robustness.md     | 312aef1d | 3 Oct, 09:52:27 | 09:53:32      | 132: 112 plus 20 battery backtests                                 |
| study2_3_robustness.md          | 068a3e03 | 4 Oct, 09:18:39 | 09:26:27      | 162: 132 plus 30 (24 as-known, 4 cap, 2 rank-weighted)             |

The run logs (`analysis/output/study2/v2_run_log.md`, `v3_run_log.md`, `study2_3_run_log.md`) hold the SHA-256 of every result table at its first run; the notebook recomputes them on each run and stops if one differs. Result files of Studies 2.1 and 2.2 keep the codes V2a to V2d inside (Appendix A).

## Study 2.1: the four variants

TABLE 28. Post-hoc variants: net Sharpe by window and test statistics at 10 bp. Every column, the test window included, is post hoc. Test figures are the test slice of a continuous run from May 2004 (A3's ledger starts from an empty book in November 2014). IC, alpha  $t$  and placebo are one-month statistics with 6 Newey–West lags; DSR counts 112 looks as independent trials, a rough deflation since the looks are correlated.

**POST-HOC**

| Variant | Research | Test  | Full  | Post-2010 | Last 18m | Test IC ( $t$ ) | Alpha $t$ | Placebo $p$ | DSR  |
|---------|----------|-------|-------|-----------|----------|-----------------|-----------|-------------|------|
| W       | +0.09    | +0.16 | +0.15 | +0.12     | −1.71    | +0.049 (+2.04)  | +0.29     | 0.02        | 0.02 |
| SC      | −0.06    | −0.15 | −0.11 | −0.16     | −1.34    | +0.042 (+1.65)  | −0.84     | 0.05        | 0.00 |
| W+MOM   | +0.27    | +0.35 | +0.32 | +0.24     | −0.78    | +0.032 (+1.21)  | +1.36     | 0.26        | 0.09 |
| HQ12    | +0.50    | +0.29 | +0.35 | +0.30     | +0.29    | −0.020 (−0.87)  | +1.37     | 0.85        | 0.06 |

The reading rule declared with the variants: positive Sharpe at 25 bp, IC  $t \geq 2$ , alpha  $t \geq 2$ , positive IC in both test halves, placebo  $p < 0.05$  and DSR above 0.95; none of the four meets it. HQ12's pre-declared one-month placebo  $p$  is 0.85; a horizon-matched placebo added after the run gives 0.06 in the test and 0.04 over the full sample.

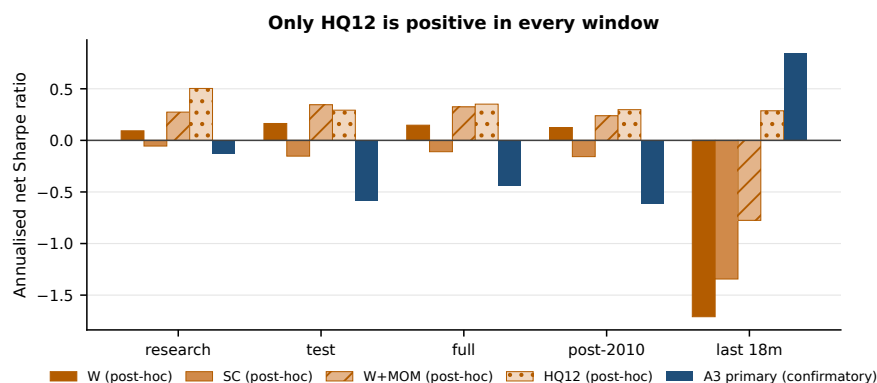


FIGURE 11. Net Sharpe by window. HQ12 is the only post-hoc variant positive in every window; W and W+MOM lost heavily in the last 18 months.

**POST-HOC** / **CONFIRMATORY** (A3 bars)

Study 2.2: the HQ12 battery on fixed-lag data

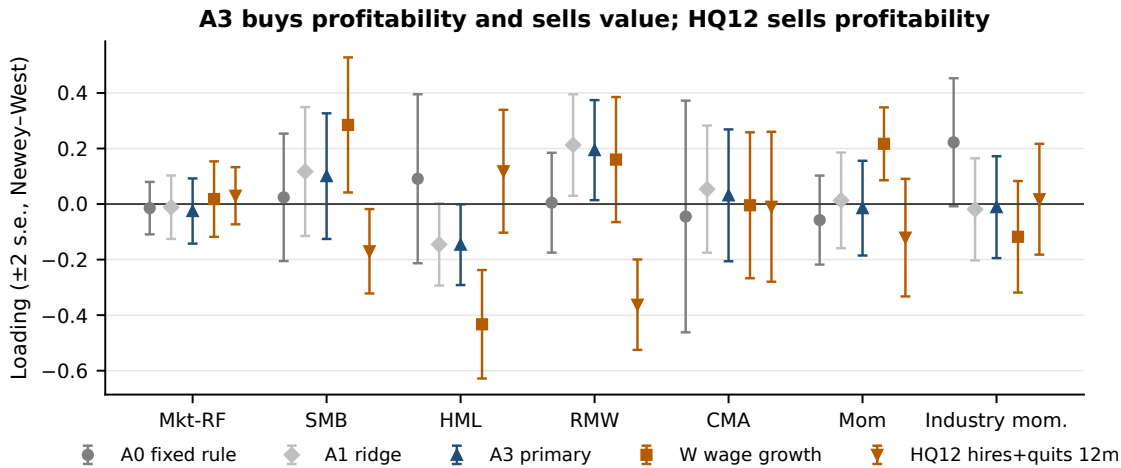


FIGURE 12. Factor loadings of the long-short books: A3 buys profitability and sells value; HQ12 sells profitability. CONFIRMATORY (A0, A1, A3) / POST-HOC (W, HQ12)

TABLE 29. R1 components: each labor flow alone, signed like HQ12 and held 12 months.

POST-HOC

| Sharpe research | Sharpe test | Sharpe full | Test IC 12m (t) | Test alpha t |
|-----------------|-------------|-------------|-----------------|--------------|
| +0.41           | +0.19       | +0.27       | +0.093 (+2.90)  | +0.79        |
| +0.18           | +0.16       | +0.15       | +0.085 (+1.75)  | +0.81        |
| +0.10           | +0.42       | +0.29       | +0.082 (+1.38)  | +1.48        |
| +0.11           | +0.16       | +0.13       | +0.032 (+0.87)  | +0.69        |
| +0.50           | +0.29       | +0.35       | +0.096 (+2.43)  | +1.37        |

TABLE 30. R2 horizon profile of the HQ12 score: IC ( $t$ ) by window.

POST-HOC

| research       | test           | full           |
|----------------|----------------|----------------|
| −0.016 (−0.59) | −0.020 (−0.87) | −0.019 (−1.08) |
| +0.005 (+0.18) | +0.007 (+0.18) | +0.005 (+0.21) |
| −0.004 (−0.11) | +0.017 (+0.40) | +0.006 (+0.20) |
| +0.023 (+0.48) | +0.069 (+1.98) | +0.046 (+1.52) |
| +0.047 (+0.90) | +0.096 (+2.43) | +0.071 (+2.06) |
| +0.038 (+0.59) | +0.085 (+2.00) | +0.059 (+1.41) |
| +0.009 (+0.11) | +0.063 (+1.86) | +0.034 (+0.72) |

TABLE 31. R3 holding period: Sharpe by window.

POST-HOC

| Holding   | research | test  | full  | post-2010 | last 18m | Test turnover |
|-----------|----------|-------|-------|-----------|----------|---------------|
| 6 months  | +0.31    | +0.05 | +0.15 | +0.05     | +0.50    | 0.49          |
| 9 months  | +0.43    | +0.14 | +0.24 | +0.16     | +0.29    | 0.40          |
| 12 months | +0.50    | +0.29 | +0.35 | +0.30     | +0.29    | 0.37          |
| 18 months | +0.58    | +0.30 | +0.38 | +0.34     | +0.31    | 0.28          |

TABLE 32. R4 sub-periods.

POST-HOC

| Window                        | Months | Sharpe | Net p.a. | IC 12m (t)     |
|-------------------------------|--------|--------|----------|----------------|
| test: first half              | 71     | −0.21  | −1.4%    | +0.097 (+1.75) |
| test: second half             | 71     | +0.64  | +6.5%    | +0.095 (+1.60) |
| test: excl. Mar 2020-Dec 2021 | 120    | +0.37  | +3.0%    | +0.105 (+2.23) |
| test: pre-2020                | 62     | +0.14  | +0.8%    | +0.114 (+1.92) |
| test: 2020 onward             | 80     | +0.37  | +3.9%    | +0.080 (+1.54) |
| full: first half              | 134    | +0.33  | +2.0%    | +0.038 (+0.73) |
| full: second half             | 134    | +0.38  | +3.3%    | +0.107 (+2.66) |
| full: excl. Mar 2020-Dec 2021 | 246    | +0.40  | +2.9%    | +0.073 (+1.93) |
| full: pre-2020                | 188    | +0.36  | +2.2%    | +0.068 (+1.57) |

TABLE 33. R5 costs: Sharpe by cost level and break-even.

POST-HOC

| Window    | 0 bp  | 10 bp | 25 bp | 50 bp | Break-even (bp) |
|-----------|-------|-------|-------|-------|-----------------|
| research  | +0.58 | +0.50 | +0.39 | +0.20 | 76              |
| test      | +0.34 | +0.29 | +0.22 | +0.09 | 68              |
| full      | +0.41 | +0.35 | +0.26 | +0.11 | 69              |
| post-2010 | +0.35 | +0.30 | +0.21 | +0.07 | 63              |
| last 18m  | +0.32 | +0.29 | +0.24 | +0.16 | 99              |

TABLE 34. R6 factor regression with 12 Newey–West lags.

POST-HOC

| Test (HAC 12)  | Full (HAC 12)  |
|----------------|----------------|
| +0.26% (+1.48) | +0.26% (+1.92) |
| +0.03 (+0.58)  | −0.00 (−0.08)  |
| −0.17 (−2.24)  | −0.02 (−0.31)  |
| +0.12 (+1.07)  | +0.10 (+1.23)  |
| −0.36 (−4.45)  | −0.16 (−1.83)  |
| −0.01 (−0.07)  | −0.09 (−0.87)  |
| −0.12 (−1.14)  | −0.03 (−0.51)  |
| +0.02 (+0.17)  | +0.06 (+0.93)  |

TABLE 35. R7 drop one group, every run.

POST-HOC

| industry              | dropped group | test sharpe | full sharpe |
|-----------------------|---------------|-------------|-------------|
| Mining & logging      | 1             | +0.33       | +0.25       |
| Construction          | 2             | +0.30       | +0.34       |
| Durable mfg           | 3             | +0.30       | +0.30       |
| Nondurable mfg        | 4             | +0.40       | +0.36       |
| Wholesale             | 5             | +0.36       | +0.37       |
| Retail                | 6             | +0.29       | +0.35       |
| Transport & utilities | 7             | +0.30       | +0.34       |
| Information           | 8             | +0.41       | +0.54       |
| Finance & insurance   | 9             | +0.23       | +0.23       |
| Real estate           | 10            | +0.48       | +0.43       |
| Prof. & business svcs | 11            | +0.26       | +0.33       |
| Health care           | 12            | +0.18       | +0.36       |
| Accommodation & food  | 13            | +0.32       | +0.28       |

TABLE 36. R8 HQ12 by tightness regime at the decision (recorded without a claim).

POST-HOC

| Months | Sharpe | Net p.a. | IC 12m (t)     |
|--------|--------|----------|----------------|
| 88     | −0.00  | −0.0%    | +0.115 (+2.52) |
| 54     | +0.66  | +6.8%    | +0.061 (+1.02) |
| 75     | +0.72  | +6.0%    | +0.103 (+2.00) |
| 67     | −0.14  | −1.2%    | +0.088 (+1.59) |
| 182    | +0.06  | +0.4%    | +0.048 (+1.19) |
| 86     | +0.83  | +7.6%    | +0.126 (+2.19) |
| 75     | +0.72  | +6.0%    | +0.103 (+2.00) |
| 193    | +0.19  | +1.4%    | +0.059 (+1.40) |

TABLE 37. R9 block-bootstrap 90% interval.

POST-HOC

| Window | Sharpe | 90% interval   | Months | Excludes 0 |
|--------|--------|----------------|--------|------------|
| test   | +0.29  | [−0.07, +0.65] | 142    | no         |
| full   | +0.35  | [+0.05, +0.64] | 268    | yes        |

TABLE 38. R11 fundamental law in the test window: IC, breadth, transfer coefficient, implied and realised ratio.

CONFIRMATORY (A3) / POST-HOC

| Book (test window)      | IC     | Horizon | BR/yr | TC   | Implied IR | Ceiling (TC=1) | Realised Sharpe |
|-------------------------|--------|---------|-------|------|------------|----------------|-----------------|
| A3 primary              | −0.021 | 1m      | 156   | 0.80 | −0.21      | −0.27          | −0.58           |
| W wage growth           | +0.049 | 1m      | 156   | 0.91 | +0.56      | +0.62          | +0.16           |
| HQ12 -(hires+quits) 12m | +0.096 | 12m     | 13    | 0.48 | +0.17      | +0.35          | +0.29           |

TABLE 39. Deflated Sharpe at 112 and 132 looks.

POST-HOC

| Window | DSR (112 trials) | DSR (132 trials) | Benchmark Sharpe (132) |
|--------|------------------|------------------|------------------------|
| test   | 0.02             |                  | 0.76                   |
| full   | 0.03             |                  | 0.56                   |
| test   | 0.00             |                  | 0.76                   |
| full   | 0.00             |                  | 0.56                   |
| test   | 0.09             |                  | 0.76                   |
| full   | 0.16             |                  | 0.56                   |
| test   | 0.06             |                  | 0.76                   |
| full   | 0.18             |                  | 0.56                   |

TABLE 40. The pre-declared reading of the battery: HQ12 is fragile on R4.

POST-HOC

| condition                                    | holds |
|----------------------------------------------|-------|
| R1 hires-only and quits-only test Sharpe > 0 | yes   |
| R2 test 12m IC t >= 2 and 1m IC t < 2        | yes   |
| R3 holding 9 and 18 months test Sharpe > 0   | yes   |
| R4 both test halves Sharpe > 0               | no    |
| R5 test break-even cost > 25 bp              | yes   |
| R7 minimum drop-one test Sharpe > 0          | yes   |
| R9 90% interval excludes 0 (test or full)    | yes   |

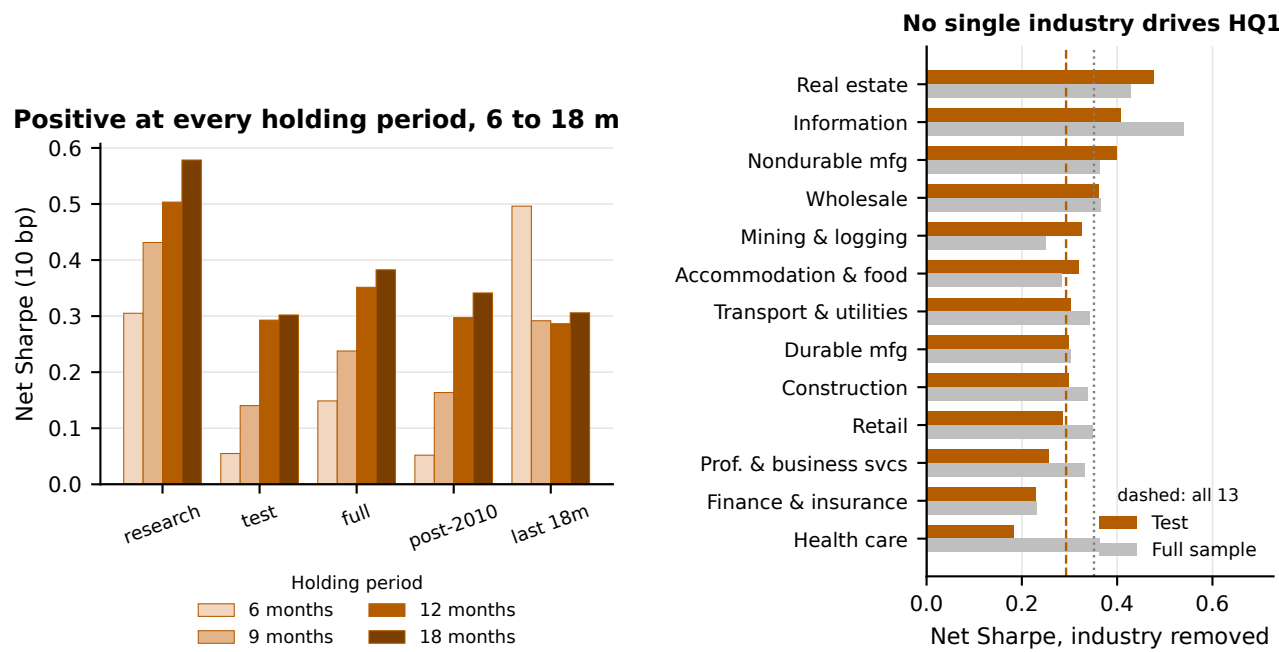


FIGURE 13. Left: HQ12 is positive at every holding period from 6 to 18 months. Right: no single industry carries HQ12; removing any one leaves its test Sharpe between +0.18 and +0.48.

POST-HOC

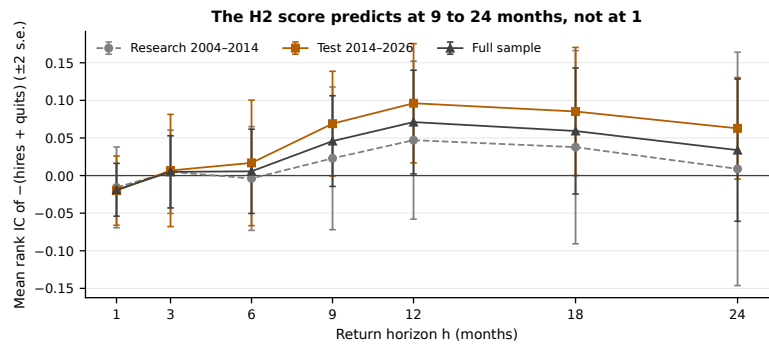


FIGURE 14. Horizon profile of the HQ12 score in every window: no one-month IC, positive from 9 to 24 months.

POST-HOC

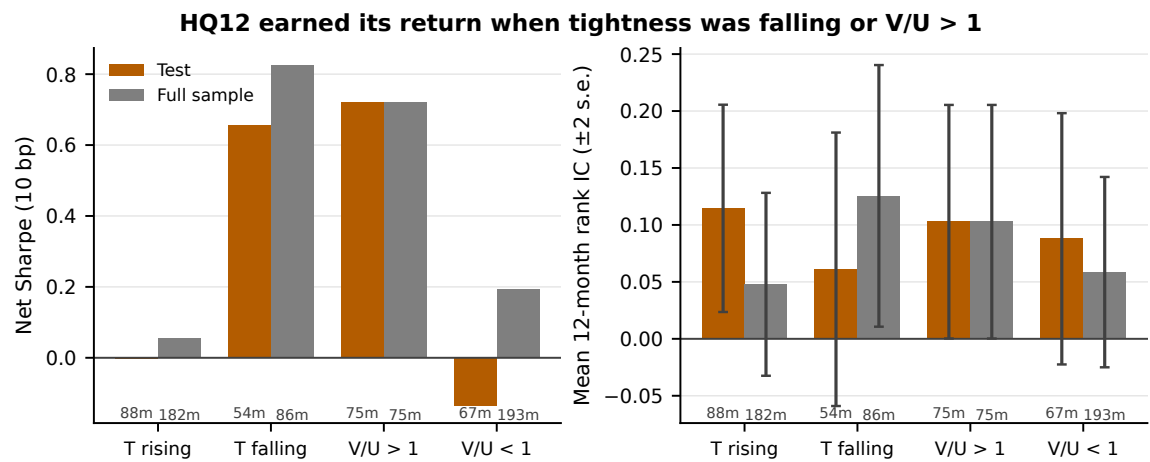


FIGURE 15. HQ12 conditional on tightness at the decision, recorded without a claim: the return came in months when tightness was falling or  $V/U > 1$ , while the 12-month IC was highest when  $T$  was rising (+0.115,  $t = 2.52$ ); 54 to 88 months per cell.

POST-HOC

### Study 2.3: as-known rerun, risk overlay, rank weights, agents, capacity

The as-known inputs come from Alex's data package. Before any variant was rerun we rebuilt his raw features from the as-known panel and his A0 score from his standardised features: both match his saved files to machine precision.

TABLE 41. R1: the four variants on fixed-lag and on as-known data, test window. The last two columns are the share of months in which both versions rank the same three groups highest and lowest. POST-HOC

| Variant (test window) | Sharpe, fixed lag | Sharpe, as known | IC (t), fixed lag | IC (t), as known | Alpha t | Same top 3 | Same bottom 3 |
|-----------------------|-------------------|------------------|-------------------|------------------|---------|------------|---------------|
| W                     | +0.16             | +0.13            | +0.049 (+2.04)    | +0.041 (+1.73)   | +0.18   | 76%        | 70%           |
| SC                    | −0.15             | +0.24            | +0.042 (+1.65)    | +0.021 (+0.75)   | +0.71   | 0%         | 0%            |
| W+MOM                 | +0.35             | +0.46            | +0.032 (+1.21)    | +0.036 (+1.40)   | +1.93   | 84%        | 74%           |
| HQ12                  | +0.29             | +0.20            | +0.096 (+2.43)    | +0.085 (+2.26)   | +0.92   | 39%        | 51%           |

TABLE 42. R1: HQ12 battery, fixed lag against as known, every row. POST-HOC

| HQ12 check                                                | Fixed lag (Study 2.2) | As known (Study 2.3) |
|-----------------------------------------------------------|-----------------------|----------------------|
| Test Sharpe (10 bp)                                       | +0.29                 | +0.20                |
| Full-sample Sharpe (10 bp)                                | +0.35                 | +0.31                |
| 12-month IC, test                                         | +0.10                 | +0.08                |
| 12-month IC t, test                                       | +2.43                 | +2.26                |
| 1-month IC t, test                                        | −0.87                 | −0.42                |
| Component hires only, test Sharpe                         | +0.19                 | +0.19                |
| Component quits only, test Sharpe                         | +0.16                 | +0.17                |
| Holding 9 months, test Sharpe                             | +0.14                 | +0.16                |
| Holding 18 months, test Sharpe                            | +0.30                 | +0.11                |
| First test half, Sharpe                                   | −0.21                 | −0.23                |
| Second test half, Sharpe                                  | +0.64                 | +0.49                |
| Test Sharpe at 0 bp                                       | +0.34                 | +0.25                |
| Test Sharpe at 25 bp                                      | +0.22                 | +0.13                |
| Test Sharpe at 50 bp                                      | +0.09                 | −0.00                |
| Break-even cost, test (bp)                                | 68                    | 50                   |
| Drop one group, minimum test Sharpe                       | +0.18                 | +0.05                |
| Drop one group, maximum test Sharpe                       | +0.48                 | +0.36                |
| Bootstrap 90% interval, test: low                         | −0.07                 | −0.14                |
| Bootstrap 90% interval, test: high                        | +0.65                 | +0.57                |
| Alpha per month (%), test, HAC 12                         | +0.26                 | +0.16                |
| Alpha t, test, HAC 12                                     | +1.48                 | +0.92                |
| Bootstrap 90% interval, full: low                         | +0.05                 | +0.02                |
| Bootstrap 90% interval, full: high                        | +0.64                 | +0.60                |
| Alpha per month (%), full, HAC 12                         | +0.26                 | +0.21                |
| Alpha t, full, HAC 12                                     | +1.92                 | +1.81                |
| Deflated Sharpe, test (132 looks fixed lag; 162 as known) | +0.05                 | +0.02                |

TABLE 43. R2: ex-ante volatility of the unit book and gross exposure needed to reach 10%: 5th, 50th and 95th percentile over the test months. POST-HOC

| book | unit vol p5 | unit vol p50 | unit vol p95 | gross needed p5 | gross needed p50 | gross needed p95 |
|------|-------------|--------------|--------------|-----------------|------------------|------------------|
| A3   | 0.045       | 0.075        | 0.110        | 1.85            | 2.60             | 4.19             |
| HQ12 | 0.033       | 0.064        | 0.103        | 1.79            | 2.45             | 4.24             |

TABLE 44. R3: top 3 / bottom 3 against rank-weighted books, test window. TC is the transfer coefficient; the implied IR is  $IC \times \sqrt{BR} \times TC$ .

POST-HOC

| Book | Weights          | Test Sharpe | TC   | Turnover | Implied IR |
|------|------------------|-------------|------|----------|------------|
| A3   | top 3 / bottom 3 | −0.58       | 0.80 | 0.91     | −0.21      |
| A3   | rank-weighted    | −0.49       | 0.83 | 0.86     | −0.22      |
| HQ12 | top 3 / bottom 3 | +0.29       | 0.48 | 0.37     | +0.17      |
| HQ12 | rank-weighted    | +0.31       | 0.49 | 0.37     | +0.17      |

TABLE 45. R4: agent proposals by study, arm and seed: valid proposals, best and mean research IC, eligible features (seed 1 only) and share with a regime gate.

POST-HOC

| Study    | Arm           | Seed | Valid | Best IC | Mean IC | Eligible | Regime-gated |
|----------|---------------|------|-------|---------|---------|----------|--------------|
| Study 1  | independent   | 1    | 17/18 | +0.101  | +0.014  | 2        | 29%          |
| Study 1  | independent   | 2    | 17/18 | +0.102  | −0.006  |          | 24%          |
| Study 1  | independent   | 3    | 13/18 | +0.046  | −0.014  |          | 15%          |
| Study 1  | communicating | 1    | 16/18 | +0.052  | −0.003  | 2        | 25%          |
| Study 1  | communicating | 2    | 17/18 | +0.058  | +0.005  |          | 6%           |
| Study 1  | communicating | 3    | 18/18 | +0.037  | −0.008  |          | 22%          |
| Study 1b | independent   | 1    | 18/18 | +0.088  | −0.002  | 1        | 28%          |
| Study 1b | independent   | 2    | 18/18 | +0.045  | −0.006  |          | 11%          |
| Study 1b | independent   | 3    | 18/18 | +0.045  | +0.002  |          | 6%           |
| Study 1b | communicating | 1    | 17/18 | +0.125  | +0.026  | 6        | 6%           |
| Study 1b | communicating | 2    | 18/18 | +0.191  | +0.049  |          | 39%          |
| Study 1b | communicating | 3    | 17/18 | +0.091  | +0.009  |          | 47%          |

TABLE 46. R4: arm gap against seed spread, exact permutation test over 20 relabelings.

POST-HOC

| Study    | Metric  | Communicating minus independent | Largest seed range | Gap within seed range | Permutation p |
|----------|---------|---------------------------------|--------------------|-----------------------|---------------|
| Study 1  | best IC | −0.034                          |                    | 0.056 yes             | 0.30          |
| Study 1  | mean IC | −0.001                          |                    | 0.028 yes             | 1.00          |
| Study 1b | best IC | +0.076                          |                    | 0.099 yes             | 0.10          |
| Study 1b | mean IC | +0.030                          |                    | 0.040 yes             | 0.10          |

TABLE 47. R5: tokens per proposal call by arm.

POST-HOC

| Study    | Arm           | Calls | Tokens, total | Tokens per call | Output tokens per call |
|----------|---------------|-------|---------------|-----------------|------------------------|
| Study 1  | independent   | 54    | 120,932       | 2,239           | 183                    |
| Study 1  | communicating | 54    | 134,185       | 2,485           | 193                    |
| Study 1b | independent   | 54    | 180,555       | 3,344           | 1,257                  |
| Study 1b | communicating | 54    | 214,003       | 3,963           | 1,509                  |

TABLE 48. R5: Spearman correlation between a proposal's output tokens and its research IC.

POST-HOC

| Study    | Proposals | Spearman, output tokens vs IC | Spearman, output tokens vs absolute IC |
|----------|-----------|-------------------------------|----------------------------------------|
| Study 1  | 98        | +0.13                         | +0.21                                  |
| Study 1b | 106       | +0.41                         | +0.24                                  |

TABLE 49. R6: participation ratio of the 13-group correlation matrices.

POST-HOC

| matrix                                                    | participation ratio |
|-----------------------------------------------------------|---------------------|
| 13 group excess returns                                   | 2.0                 |
| 13 group relative returns (group minus equal-weight mean) | 9.1                 |
| z(W) across groups                                        | 7.4                 |
| HQ12 score across groups                                  | 10.1                |

TABLE 50. R7: capacity with the proxy ETFs: the binding ETF, its average daily dollar volume, and the assets at which monthly trading reaches 1% and 5% of it. POST-HOC

| Book | Binding ETF | Its daily volume (USD m) | Monthly trade (share of assets) | Assets at 1% of a day (USD m) | Assets at 5% (USD m) | SPY daily volume (USD bn) |
|------|-------------|--------------------------|---------------------------------|-------------------------------|----------------------|---------------------------|
| A3   | PEJ         | 3.2                      | 6.3%                            | 0.5                           | 2.5                  | 47                        |
| HQ12 | PEJ         | 3.2                      | 2.7%                            | 1.2                           | 5.9                  | 47                        |

TABLE 51. R8: judge with a verbatim quote on the 25 hand-labelled pairs. “Defined” counts JSON-only answers; the third row of each study accepts JSON wrapped in a code fence and is outside the pre-declared rule. POST-HOC

| Study    | Judge                                            | Pairs | Usable | Agree with human | Judge yes | Human yes | False yes | Missed yes |
|----------|--------------------------------------------------|-------|--------|------------------|-----------|-----------|-----------|------------|
| Study 1  | original judge                                   | 8     | 2      | 1                | 1         | 1         | 1         | 0          |
| Study 1  | judge with verbatim quote                        | 8     | 8      | 7                | 2         | 1         | 1         | 0          |
| Study 1  | with quote, fenced JSON accepted (supplementary) | 8     | 8      | 7                | 2         | 1         | 1         | 0          |
| Study 1b | original judge                                   | 17    | 17     | 10               | 10        | 3         | 7         | 0          |
| Study 1b | judge with verbatim quote                        | 17    | 7      | 3                | 5         | 3         | 4         | 0          |
| Study 1b | with quote, fenced JSON accepted (supplementary) | 17    | 17     | 9                | 11        | 3         | 8         | 0          |

TABLE 52. R9: feedback ablation, independent arm, research data only, with and without the tightness-tercile split in the feedback. “Informative” counts months in which a proposal's cross-section is not constant. POST-HOC

| Feedback              | Seeds | Proposals | Valid | With a regime gate | Share | Informative in under half of months | Best research IC |
|-----------------------|-------|-----------|-------|--------------------|-------|-------------------------------------|------------------|
| with tercile split    | 3     | 54        | 48    | 4                  | 8%    | 8%                                  | +0.070           |
| without tercile split | 3     | 54        | 50    | 3                  | 6%    | 6%                                  | +0.070           |

The ablation rebuilt the agents’ research inputs from Alex’s package; they reproduce 94 of the 98 saved feedbacks exactly, and the largest difference is 0.11 in a  $t$ -statistic. The judge calls for Study 1b pairs used a newer command-line client than the Study 1 pairs because the older one does not serve that model; calls that failed in transport were repeated once, and every call is in the log (analysis/output/study2/11m/).

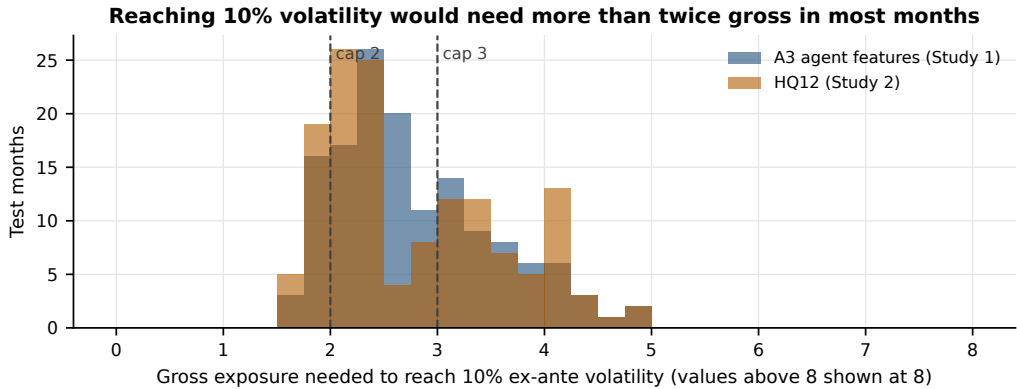


FIGURE 16. Reaching 10% volatility would need more than twice gross in most test months, so the cap of 2 binds. POST-HOC

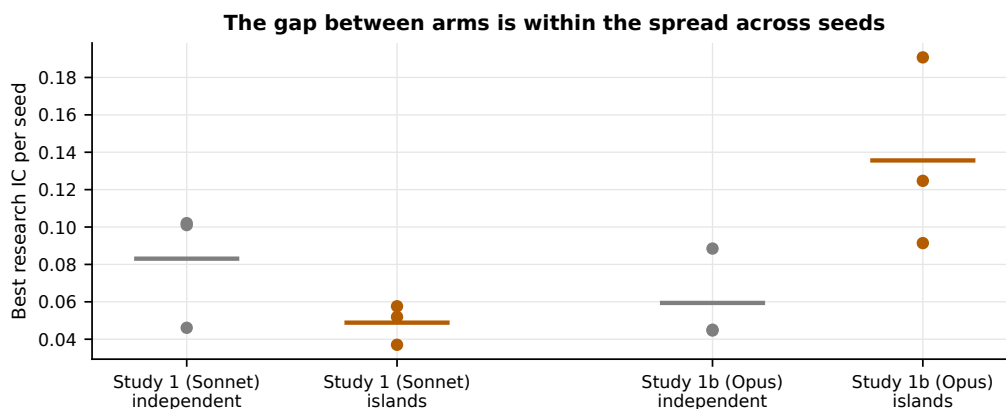


FIGURE 17. Best research IC per seed: the gap between arms is within the spread across seeds in both studies.

POST-HOC

## D Forward test: frozen specification and Q4 2026 books

- `analysis/specs/study2_1_variants.md`: four post-hoc variants and a reading rule; SHA-256 `ca658234...`, frozen 2026-10-03T09:00:13Z, first run 09:01:55Z.
- `analysis/specs/forward_spec.md`: decision 30 September 2026, information cutoff 29 September, one tranche held to 31 December, 60-month beta hedge in SPY, 10 bp and 2 bp costs; SHA-256 `735a8ffa...`, frozen 09:03:56Z.
- `analysis/specs/forward_amendment_001.md`: the fixed-lag information set (JOLTS July 2026, CES August 2026, as published on 29 September) after the pre-committed check stopped the first run; W+MOM and HQ12 added after the Study 2.1 results were seen and before any Q4 return beyond 1–2 October; SHA-256 `c76cde86...`, frozen 09:16:11Z; book built 09:17:55Z.
- `analysis/specs/study2_2_hq12_robustness.md`: the HQ12 battery; SHA-256 `312aef1d...`, frozen 09:52:27Z, first run 09:53:32Z. `analysis/specs/study2_3_robustness.md`: SHA-256 `068a3e03...`, frozen 4 October 09:18:39Z, before its first output.
- Books: `analysis/output/forward/book_2026Q4.csv` (group weights for W, SC, W+MOM and HQ12), `book_2026Q4_etf.csv` (netted ETF weights), `book_2026Q4_hedge.csv` (SPY hedge per unit:  $-0.40$ ,  $-0.06$ ,  $-0.24$ ,  $-0.02$ ), `etf_proxies.csv`. Holdings: W is long construction, durable manufacturing and information and short mining, health care and accommodation; SC is long durable manufacturing, retail and transport and short real estate, health care and accommodation; W+MOM is long construction, durable manufacturing and finance and short retail, transport and professional services; HQ12 is long finance, real estate and accommodation and short mining, durable manufacturing and retail. W+MOM's tradable book has four legs, because its long durable-manufacturing and short professional-services positions both map to the industrials ETF and cancel.
- Power: HQ12's test volatility is about 9% a year, so one quarter's return has a standard deviation near 5%; a Q4 return between  $-10\%$  and  $+10\%$  is within two standard deviations of zero. The book was first built at 09:17:55Z on 3 October: commit `3e8ecd7` (09:18:29Z) holds a byte-identical `book_2026Q4.csv` (SHA-256 `857275...`); `book_first_build.sha256` records that first build and later notebook runs must reproduce it.
- Evaluation: the ETF books from the 30 September close to the 31 December close after costs, and the same books on Ken French group returns once published; reported whatever they show. Cadence: a weekly tracker was considered and deferred by team decision – with one quarter of data and no capital at risk, a weekly readout adds noise without adding power (Section 3.3, Looks, forward test and limits). We'll evaluate once at the 31 December close.

## E Claude, interaction by interaction

---

Condensed from the team’s written evaluation (Al Yazid Bensaid, 3 October 2026). Paths are relative to `study1_preregistered/outputs/follow_the_workers/` unless stated; Study 1b has the same files under `follow_the_workers_opus55_xhigh/`.

Claude took every AI role: Sonnet 5.5 for Study 1, Opus 5.5 at extra-high effort for Study 1b. This term’s course allows it; the switch from the ChatGPT template is logged as preregistration amendments 001 and 002 (`outputs/prereg_amendments/`, `outputs/deviations.md`), and the log file keeps the template name `chatgpt_log.md` with Claude content.

*P1, data and code audit.* The system prompt made Claude “a careful data and code auditor” with no tools; the prompt in Section 2.3 was followed by the manifest, the crosswalk and the as-of code (38,717 characters). It returned 25 numbered findings, all tested (`outputs/p1_review.md`): five were claims the files did not support, among them that group 10’s CES series “silently vanishes” (584 shared, nonmissing rows were present); one led to a metadata fix (the crosswalk now separates the prescribed match from the verified French scope); the rest were documented design choices or limitations, and none was a look-ahead error.

*P2, signal agents.* Six agents (three independent, three communicating) over three seeds proposed six features each in a 14-operation grammar, blind to names and dates, with research-period feedback only. 98 of 108 proposals passed validation; the blinding audit of all 199 calls passed (108 proposals, 18 migration reports, one syntax repair). The communicating agents exchanged migration reports capped at 120 words; 11 of 18 were rejected, 10 of them for running to 121–133 words although the prompt asked the agent to check its count. Selection was mechanical, two per arm from seed 1 only, and no engineer edited a proposal. Both independent-arm picks, decoded after the test, were low-tightness gates: `regime(yoy(Earnings), "low")` and `regime(spread(tsz(Layoffs, 60), tsz(Hires, 60)), "low")`, active in 3 of 142 test months. The agents optimised what they were shown: IC split by tightness tercile. Blinded, they could not know that  $T$  would stay high after 2014. Next time, keep regime splits out of agent feedback, or require a minimum share of active months in research. (We tested the first suggestion in Study 2.3; see Section 3.4 and Table 52.)

*P3, judge.* In Study 1b, items 2, 3 and 13 cite “submission 2’s significant negative IC”, a result that exists in another report; item 12 cites “submission 4” from a report with two submissions. Items 8, 16 and 17 are judgment calls (a methodological tip reused; a report’s topic shared but its result contradicted). The same proposal paired with two reports (items 12 and 17) got a yes both times, which is what a judge that matches on vocabulary would do. With 17 dependent pairs the sample is small. In Study 1 only 8 of 36 judge outputs were usable and 8 pairs were audited. The judge said “uses a claim” 10 times where the human said 3, so its 62% uptake rate overstates communication between agents.

*P4, red team.* Six of the ten concerns were already covered by registered diagnostics (placebo, 25 bp costs, test halves, tightness, data versions); for the other four no additional test was adopted. The response asserted a permutation  $p$ -value “already” above 0.10 that nobody had computed, and proposed a 95th-percentile gate, a DSR of 0.95 and a rolling 24-month threshold after seeing the research tables.

*P5, post-mortem review.* Written from the test results, so its fixes are post hoc by construction; the review’s own errors (a vendor guess from a code path, a momentum definition that included the latest month, wage ICs from a non-frozen start date) are recorded with their corrections in `docs/FINDINGS_AND_PROPOSAL.md`.

*Bottom line.* Every Claude output needed a human check: 25 P1 dispositions, 10 P4 dispositions and 25 hand-labelled migration pairs (8 in Study 1, 17 in Study 1b).

## E.1 Transcripts and verbatim excerpts

- P1: prompt `outputs/p1_request.json`; response `outputs/p1_response.md`; dispositions `outputs/p1_review.md`: 25 findings, 5 not supported by the files, 1 metadata fix, 19 documented design choices, limitations or checks that found no error.
- P2: system and turn prompts in `params.py` (`SYSTEM_PROMPT`, `TURN_PROMPT`); every call with its raw response in `outputs/agents/log.jsonl`; proposals in `outputs/candidates.csv`; selected features decoded in `analysis/output/study1/tables/d7_selected_features.csv`.
- P3: migration reports in `outputs/agents/migrations.json`; the judge prompt in `agents.py`, function `trace_migrations`; judgments in `outputs/agents/migration_judgments.json`; the human audit in `outputs/human_review_submission.json` (Study 1b also has a `.md` version).
- P4: prompt `outputs/p4_request.json`; response `outputs/p4_response.md`; dispositions `outputs/p4_review.md`.
- P5: the review and its output, `docs/FINDINGS_AND_PROPOSAL.md`, with the numbers in `analysis/output/study1/diagnostics.md`.
- P6: task prompts for each stage of the iteration (rebuild and diagnostics; Study 2.1 and 2.2 freezes, forward test and report draft; repository passes; Study 2.3 and this rewrite) and a one-page rules file (Study 1 folder read-only, one label per result, specs hashed before they run, a source comment on every number, stop at a failed gate), kept outside the repository; outputs in `analysis/run.ipynb`, `analysis/output/`, `results/` and this report.
- Study 2.3 calls (judge with a quote; feedback ablation): every prompt and raw response in `analysis/output/study2/llm/r8_calls.jsonl` and `r9_calls.jsonl`; code in `analysis/llm_runs.py`.

*Verbatim excerpts.* Four short excerpts, copied from the files above without editing.

### P1 OUTPUT, OPENING OF THE RESPONSE (`OUTPUTS/P1_RESPONSE.MD`)

Scope: only the supplied material. I did not verify live FRED data. I mark each item as confirmed (visible in the text) or as a question or limitation. Confirmed errors: mapping and manifest. 1. Manifest rows for CEU5500000007 and CEU5500000008 with group 10 duplicate the group 9 rows. The same series ID is mapped to two groups, and group 10 (Real estate) has no distinct CES series. [...] Test: group the manifest by id and assert one group per series. Compare the feature columns of groups 9 and 10, which will be identical.

### P2 OUTPUT, ONE AGENT PROPOSAL, DECODED AFTER THE TEST (INDEPENDENT ARM, SEED 1, TEMPORAL AGENT, PROPOSAL 3)

Expression regime(`yoy(V05)`, "low"). Hypothesis: groups with higher year-over-year change in V05 (dollars per hour) will have higher next-month returns in low-T regimes, because accelerating compensation signals stronger conditions not yet fully priced in. The effect should vanish in high T. [V05 was later decoded as average hourly earnings; the feature was selected for A3 and was active in 3 of 142 test months.]

### P3 JUDGE VERSUS HUMAN, TWO AUDITED PAIRS (`OUTPUTS/HUMAN_REVIEW_SUBMISSION.MD`, STUDY 1B)

Item 5, human yes, judge yes: "The report found that V02 outpacing V03 led to significant underperformance, and that the V03-favoring reversal was positive across T regimes. The proposal uses that V03/V02 composition signal without conditioning on T." Item 2, human no, judge yes: "It says it reverses 'an observed negative signal,' but this report has no negative signal: the V02/V03 ratio had IC 0.01 and the V02-vs-V04 shift had IC 0."

#### P4 OUTPUT, FIRST OF TEN REASONS (OUTPUTS/P4\_RESPONSE.MD)

1. The IC signals are pure noise (no predictive content). All t-statistics are below 1.3 and the signs flip across variants. Test: a block-bootstrap or permutation test of IC, shuffling signals within each date, 1000 or more draws, with Newey-West lags of 6. Do not implement if: the permutation p-value is above 0.10 for any variant. This is already the case, so the answer is do not implement. [Disposition in p4\_review.md: covered by the registered circular-shift placebo; no new test adopted.]

## F Key code

*As-of lookup.* The value of a series known on a cutoff date (Study 1 data.py).

```
def vintage_view(frame, cutoff, first=False):
    cutoff = pd.Timestamp(cutoff).strftime('%Y-%m-%d')
    if first:
        eligible = frame.drop_duplicates('observation_month', keep='first')
        eligible = eligible.loc[eligible.realtime_start <= cutoff]
    else:
        eligible = frame.loc[(frame.realtime_start <= cutoff) & (frame.realtime_end >= cutoff)]
    if eligible.duplicated('observation_month').any():
        raise ValueError('Overlapping vintage intervals')
    return eligible.set_index('observation_month')[['value', 'realtime_start']].sort_index()
```

*Signal construction.* Fixed-lag signals and tightness (analysis/data.py).

```
def signals(folder=P.FRED, last_decision=None, groups=None):
    panel = labor_panel(folder)
    avg3 = lambda x: x.rolling(3, min_periods=3).mean()
    raw = {}
    to_decision = lambda frame, lag: frame.set_axis(frame.index + lag) # observation o ->
    decision o+lag
    for i, m in enumerate(P.JOLTS_MEASURES, 1):
        x = avg3(panel[m])
        raw[f'F{i}'] = to_decision(x - x.shift(12), P.JOLTS_LAG)
    h, e = avg3(panel['H']), avg3(panel['E'])
    raw['F5'] = to_decision(np.log(h / h.shift(12)), P.CES_LAG)
    raw['W'] = to_decision(np.log(e / e.shift(12)), P.CES_LAG)
    # last decision month at which every feature has its lagged observation
    end = last_decision or min(v.dropna(how='all').index.max() for v in raw.values())
    decisions = pd.period_range(P.SIGNAL_START, end, freq='M')
    features = {}
    for name, value in raw.items():
        value = value[list(groups)] if groups is not None else value # drop-one-industry
        # diagnostics
        z = standardize(value.reindex(decisions))
        z.index = z.index + 1 # earning month
        features[name] = z
    # T at decision d uses the latest month <= d-2 with both V and U observed (frozen rule); this
    # matters once: the Oct 2025 household survey (UNEMPLOY) was never collected.
    ratio = np.log(panel['V'] / panel['U']).ffill()
    t = to_decision(ratio, P.JOLTS_LAG).reindex(decisions)
    t.index = t.index + 1
    return features, t
```

*Backtest core.* Tranche, hedge, scaling and costs (Study 1 data.py, excerpt).

```
for month, score in scores.iterrows():
    tranche = pd.Series(0., index=groups)
    if signal_available.loc[month]:
        rank = score.sort_index().sort_values(kind='stable')
```

```

        tranche.loc[rank.index[-P.PORTFOLIO['longs']:]] = 1/P.PORTFOLIO['longs']
        tranche.loc[rank.index[:P.PORTFOLIO['shorts']]] = -1/P.PORTFOLIO['shorts']
    tranches.append(tranche)
    book = sum(tranches[-holding:])/holding
    betas = pd.Series({g:np.cov(historical[g],market,ddof=1)[0,1]/variance for g in groups})
    raw = np.r_[book.to_numpy(),-float(book@betas)]
    covariance = LedoitWolf().fit(historical.to_numpy()).covariance_
    risk = np.sqrt(max(0,float(raw@covariance@raw))*12)
    if risk<=0 and np.abs(raw).sum(>)>1e-14:
        raise ValueError('Degenerate portfolio risk estimate')
    scale = min(P.PORTFOLIO['vol_target']/risk,P.PORTFOLIO['gross_cap']/np.abs(raw).sum()) if
risk>0 else 0.
    weights = pd.Series(raw*scale,index=pretrade.index)
    assert abs(weights[groups]@betas+weights['market'])<1e-10
    assert weights.abs().sum()<=P.PORTFOLIO['gross_cap']+1e-10
    turnover = float((weights[groups]-pretrade[groups]).abs().sum())
    hedge_turnover = abs(weights['market']-pretrade['market'])
    transaction = turnover*cost_bps/10000+hedge_turnover*P.PORTFOLIO['hedge_cost_bps']/10000
    # Charge borrowed securities plus negative residual cash, at the stated annual rate.
    borrow_base = -weights.clip(upper=0).sum()
    finance_base = max(0,float(weights.sum()-1))
    financing = (borrow_base+finance_base)*borrow_bps/10000/12
    gross = float(weights[groups]@returns['excess'].loc[month,groups]+weights['market']*factors
['Mkt-RF'])
    net = gross-transaction-financing

```

## G Data sources and links

- Ken French Data Library: [https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data\\_library.html](https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html) (49 industry portfolios, five factors, momentum; CRSP build 202608). Live, checked 3 October 2026.
- JOLTS by industry, not seasonally adjusted: FRED series JTU&ltcode>JOR/HIR/QUR/LDR, for example <https://fred.stlouisfed.org/series/JTU2300HIR> (checked 3 October 2026, series runs through Aug 2026). Codes 110099, 2300, 3200, 3400, 4200, 4400, 480099, 5100, 5200, 5300, 540099, 6200, 7200.
- CES hours and earnings: CEU&ltcode>07 and CEU&ltcode>08 (construction earnings CES20000000008); codes 10000000, 20000000, 31000000, 32000000, 41420000, 42000000, 43000000, 50000000, 55000000, 60000000, 65000000, 70000000. Checked 3 October 2026 via FRED.
- Tightness: JTSJOL and UNEMPLOY. Vintages: <https://alfred.stlouisfed.org/> (live, checked 3 October 2026).
- As-known panels and saved outputs of Study 1 and Study 1b: Alex Roesler's data package (4 October 2026; 1,191 files checked against its inventory), read from `analysis/data/alex_package/`.
- ETF proxies and their caveats: `analysis/output/forward/etf_proxies.csv`. Daily dollar volume for the capacity note: Yahoo Finance, 252 trading days to 30 September 2026.

## H From 49 French industries to 13 labor groups

TABLE 53. The 13 groups: JOLTS code, Ken French members and the largest member's share of August 2026 market capitalisation.

CONFIRMATORY

| Group                              | JOLTS  | Ken French industries                                                                        | Largest member |
|------------------------------------|--------|----------------------------------------------------------------------------------------------|----------------|
| Mining and logging                 | 110099 | Gold, Mines, Coal, Oil                                                                       | Oil 79%        |
| Construction                       | 2300   | Cnstr                                                                                        | 100%           |
| Durable manufacturing              | 3200   | Toys, BldMt, Steel, FabPr, Mach, ElcEq, Autos, Aero, Ships, Guns, Hardw, Chips, LabEq, MedEq | Chips 60%      |
| Nondurable manufacturing           | 3400   | Food, Soda, Beer, Smoke, Hshld, Clths, Drugs, Chems, Rubbr, Txtls, Paper                     | Drugs 60%      |
| Wholesale                          | 4200   | Whsl                                                                                         | 100%           |
| Retail                             | 4400   | Rtail                                                                                        | 100%           |
| Transport and utilities            | 480099 | Trans, Util                                                                                  | Util 61%       |
| Information                        | 5100   | Telcm                                                                                        | 100%           |
| Finance and insurance              | 5200   | Banks, Insur, Fin                                                                            | Banks 51%      |
| Real estate                        | 5300   | REst                                                                                         | 100%           |
| Professional and business services | 540099 | BusSv                                                                                        | 100%           |
| Health care                        | 6200   | Hlth                                                                                         | 100%           |
| Accommodation and food             | 7200   | Meals                                                                                        | 100%           |

Groups 9 and 10 share one CES series, and groups 7, 12 and 13 use broader CES supersectors. The participation ratios of the 13-group correlation matrices put a number on the effective width: 2.0 for excess returns (the market dominates), 9.1 for relative returns, 7.4 for  $z(W)$  and 10.1 for the HQ12 score.

## I Where each requirement is answered

TABLE 54. The brief and the rubric, line by line.

| Requirement                                                                                           | Where                                                                                                   | Page |
|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|------|
| 1 Executive summary: one paragraph; proposed strategy stated; “Do not implement” with justification   | Section 1                                                                                               | 1    |
| 2a Ideas: origin of the strategy (macro drivers, IC-based alphas, momentum)                           | Section 2.1, Figure 1, Table 1                                                                          | 1–2  |
| 2b Data: sources described (Ken French, FRED, BLS)                                                    | Section 2.2, Table 2; links in Appendix G                                                               | 2    |
| 2c Implementation: risk modeling                                                                      | Section 2.3, “Risk modeling”, Table 4; portfolio equation in Appendix B                                 | 4    |
| 2c Implementation: turnover and costs                                                                 | Section 2.3, “Turnover and costs”, Table 5                                                              | 4    |
| 2c Implementation: at least three high-quality prompts                                                | Section 2.3, three boxed prompts (idea generation, coding support, robustness design)                   | 5–6  |
| 3a Style and risk factor exposures                                                                    | Section 3.1, Table 7; Figure 12                                                                         | 7    |
| 3b Time patterns: full sample, post-2010, recent 12–18 months                                         | Section 3.2, Table 6 (last three columns), Figure 4                                                     | 7    |
| 3c Robustness and limits                                                                              | Section 3.3, Table 8, Table 9, Figure 3                                                                 | 7    |
| 3d AI evaluation: usefulness, flaws, adaptations                                                      | Section 3.4, Table 10                                                                                   | 9    |
| 4 Appendices: AI transcripts                                                                          | Appendix E, Appendix E.1                                                                                | 28   |
| 4 Appendices: tables and plots                                                                        | Appendix B, Appendix C                                                                                  | 11   |
| 4 Appendices: code snippets                                                                           | Appendix F                                                                                              | 30   |
| Thesis (40%): well motivated, consistent with the data, clearly explained                             | hypotheses in Section 2.1; “Answer to the thesis” box; Table 1                                          | 1    |
| Execution (40%): risk control                                                                         | risk model, hedge, volatility target and cap; Table 4; rank weights                                     | 4    |
| Execution: horizon analysis                                                                           | Figure 3; “Five things went wrong”, Horizon (p. 6); Table 9 holding periods                             | 8    |
| Execution: robustness checks                                                                          | Table 8, Table 9, placebo, bootstrap, DSR, as-known rerun                                               | 7    |
| Execution: critical evaluation of the AI’s role                                                       | Table 10; judge and feedback tests; reflection                                                          | 9–10 |
| Originality (20%): novel data use                                                                     | point-in-time JOLTS and CES vintages by industry; four data versions                                    | 2    |
| Originality: prompt design                                                                            | blinded agents in two arms with a fixed grammar; human-audited judge                                    | 5    |
| Originality: unique insights                                                                          | labor flows as slow cost news (H2, post hoc), p. 8; a quote requirement does not fix an LLM judge, p. 9 | 8    |
| AI requirement 1: three substantial interactions (idea generation, coding support, robustness design) | P2, P1, P4 in Section 2.3; six in Table 10                                                              | 5–6  |
| AI requirement 2: each documented with prompt, summarised output, critical evaluation                 | Table 10; Appendix E                                                                                    | 9    |
| AI requirement 3: reflection on strengths and limits                                                  | Section 3.4, “Reflection”                                                                               | 10   |

The brief names ChatGPT; this term’s course allows Claude, which we used in every AI role (Section 3.4).